

Risk Metrics and Fine Tuning of High Frequency Trading Strategies[☆]

Forthcoming in Mathematical Finance

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Abstract

We propose risk metrics to assess the performance of High Frequency (HF) trading strategies that seek to maximize profits from making the realized spread where the holding period is extremely short (fractions of a second, seconds or at most minutes). The HF trader maximizes expected terminal wealth and is constrained by both capital and the amount of inventory that she can hold at any time. The risk metrics enable the HF trader to fine tune her strategies by trading off different metrics of inventory risk, which also proxy for capital risk, against expected profits. The dynamics of the midprice of the asset are driven by information flows which are impounded in the midprice by market participants who update their quotes in the limit order book. Furthermore, the midprice also exhibits stochastic jumps as a consequence of the arrival of market orders that have an impact on prices which can give rise to market momentum (expected prices to trend up or down). The HF trader's optimal strategy incorporates a buffer to cover adverse selection costs and manages inventories to maximize the expected gains from market momentum.

Keywords: Algorithmic Trading; High Frequency Trading; Momentum Trading; Market Impact; Adverse Selection; Risk Metrics; Inventory Risk

1. Introduction

Computerized trading has revolutionized the way in which financial markets work. Nowadays, all major exchanges operate on electronic platforms where market participants trade using powerful hardware and customized software. A general class of computerized trading is known as Algorithmic Trading (AT) which refers to algorithms that are designed to make trading decisions and to manage inventories without human intervention. It is reported that over 60% in the US, and over 40% in Europe, of the trading volume in equities is due to AT; a figure that is expected to increase, CFTC and SEC (2010). Within the AT class

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there is a more narrowly defined set of strategies, known as High Frequency (HF) trading, that relies on ultra-high speed to process information in order to post, cancel, or amend buy and sell orders; and where inventories are managed to ensure that no positions are carried overnight after markets close (see Cvitanic and Kirilenko (2010)).

Setting up a proprietary HF trading desk or becoming an HF trader requires high initial sunk costs, but more importantly, it requires skills to develop strategies and maintain them so that they stay competitive and profitable (see Cartea and Penalva (2011)). Thus, understandably, there is a great deal of secrecy around what is it that HF traders do and how they do it. Although for some electronic markets it is possible to obtain detailed information of all messages sent to the limit order book (LOB), these publicly available data do not include trader identity, hence it is very difficult to: reverse engineer HF strategies, measure HF traders' trading profits, or assess the impact that HF trading has on the quality of the markets.

Furthermore, an important state variable which is impossible to track with publicly available data is the inventories that HF traders hold. The exception in the literature are two studies that have access to the audit-trail tape which includes the identity of each trader submitting orders to the LOB (Kirilenko et al. (2010) and Brunetti et al. (2011)). Brunetti et al use audit-trail data for all transactions in the September 2009 E-mini S&P 500 futures contract during the month of August 2009. Over that period they find that only 11 HF traders are active in the market (out of 20,000 traders) and find that the net position (aggregate inventory) of these HF traders starts at zero at the beginning of the trading day, ends at zero at the end of the trading day, and it is always between -3,000 and 3,000 contracts.¹

The picture that emerges from the current literature on HF trading is that profits are made by turning over positions over extremely short periods of time, where speed, which is paramount to process information and to make trading decisions, is at the heart of their competitive advantage. Another, much less explored trait of HF trading, is how are inventories managed. Apart from the fact that HF traders aim at holding no inventories at the end of the trading day, very little is known about how HF traders manage the accumulated positions whilst both maximizing expected profits and obeying risk controls on their inventory exposure.

In this paper we develop an HF trading strategy to post limit buy and sell orders which are continuously updated based on the dynamics of the midquote price of the asset, the shape of the LOB, and the trader's accumulated inventories. The HF strategy results from solving an optimal control problem where the HF trader maximizes expected terminal wealth whilst optimally managing inventories and by taking into account trades where she might be adversely selected. To this end, the optimal limit orders include a buffer to offset losses that arise from trading with counterparties who are better informed about the direction of subsequent price moves or losses that result from trades that have a permanent (and adverse) price impact. Moreover, inventories are continuously managed to ensure that at any point in time the strategy does not build large positions and to prevent triggering unfavorable price movements when positions are

¹The positions held by the HF traders are very low compared to those of the other traders. For instance, buyers and sellers start the day with zero inventories and by the end of the day they have accumulated around 30,000 and -20,000 contracts respectively.

unwound at the end of the trading day.

Early work on optimal postings by a securities dealer is that of Ho and Stoll (1981). More recently Avellaneda and Stoikov (2008) study the optimal HF submission strategies of bid and ask limit orders where the authors perform a small inventory asymptotics in a symmetric model and assume that market orders arrive at Poisson times and the fundamental price is independent of order arrivals. In general, very little is known about the details of the strategies that are employed by AT desks or the more specialized proprietary HF trading desks. Algorithms are designed for different purposes and to seek profits in different ways, Bouchard et al. (2011). For example, there are algorithms that are designed to find the best execution prices for investors who wish to minimize the price impact of large buy or sell orders, Bertsimas and Lo (1998), Almgren (2003), Kharroubi and Pham (2010) and Bayraktar and Ludkovski (2012). Moreover, there are HF strategies that specialize in arbitraging across different trading venues, and that seek to profit from market making and from short-term deviations in stock prices, Cartea and Jaimungal (2010), Guilbaud and Pham (2011), Cartea et al. (2011) and Guéant et al. (2011).

In this paper we assume that the HF trader maximizes expected terminal wealth and is inventory and capital constrained. The inventory constraint is a consequence of the time scale over which the HF strategies are designed to profit from buying and selling assets. HF traders use their superior speed to process information and act ahead of other slower traders. For example, some of these strategies include: arbitraging the same stock in different markets, thus absolute speed is all that matters; or to trade quicker than other market participants to take advantage of short-lived price deviations from the fundamental value of the asset. Admittedly, there are a great deal of HF strategies and all we know is that their success depends on being able to profit from roundtrip trades (buy followed by a sell or vice-versa). Therefore, because HF traders' competitive edge is speed, their strategies seek opportunities to enter and exit the market very quickly (seconds or minutes) and, as a result, holding periods are extremely short. In other words, speed cannot be employed only to execute one leg of a roundtrip trade and wait until there is an opportunity to close the position. On the contrary, HF traders only take positions that present a high chance of being closed very quickly and at a profit. Consequently, HF traders' inventories should never be too 'far' away from the optimal inventory level. The profitability of strategies that operate over "long-time" intervals depends on more attributes than just speed and, for that reason, HF traders lose their competitive advantage to more traditional professional traders.

In addition, capital requirements also constrain the amount of assets the HF trader is willing (or allowed) to hold at any point in time. For example, if the HF trader builds a large position in a stock, she will need more capital to be able to hold the position; whether required by the exchange or by her own risk controls. Similarly, if at the end of the trading day the HF trader has a large position to be carried overnight, she not only faces the risk that the market will move against her before trading commences the following business day, but the exchange will require collateral. Finally, traders know that unwinding large positions is costly because their own trades may trigger adverse price movements and this effect might be exacerbated at the end of the trading day.

Therefore, here we assume that the control problem solved by the HF trader maximizes expected terminal wealth where inventories enter in the optimization program in three different ways. First, the amount of assets that the HF trader can hold at any point in time is between an upper and lower bound. Second,

deviations of inventories from zero along the entire path of the strategy, i.e. throughout the whole trading horizon, are penalized. And third, at maturity (at most one trading day for HF traders) the inventory is liquidated at worse prices than the best available quotes because HF traders have to execute aggressive market orders to ensure that no positions are carried over after the market closes. The use of market orders is costly because there is an exchange fee for taking liquidity as well as a high chance of causing adverse price changes when there is not enough depth at the best quotes; the HF will receive less (pay more) than the best bid (best offer) when she unwinds her positions at maturity.

Furthermore, our paper contributes to the literature in two more directions. First, we propose risk metrics which are designed for HF trading strategies. Current off-the-shelf risk metrics are intended for the more traditional investment strategies and fail to provide the right decision making tools and risk controls for HF trading. Risk metrics for HF trading should encapsulate information on risk exposures that derive from risk profiles over long (hours) and short (seconds or minutes) time scales. We propose risk metrics which are based on the path of inventories throughout the entire period of the trading strategy. Our risk metrics embed information about: the lifetime for every inventory level; how quickly inventories return to their optimal level; how much time is the strategy exposed to undesired levels of inventories; and what is the exposure to terminal inventories. Thus, our risk metrics allow HF traders to fine tune their strategies to obey risk controls and to trade off inventory risk against expected profits.

Second, in our model the innovations in the fundamental value of the asset are driven by information flows and the arrival of market orders. Under normal market circumstances, information is gradually impounded in the fundamental price of the asset by market participants who are continuously updating their quotes in the LOB. Market orders, on the other hand, cause prices to jump up or down by a random amount depending on the arrival of news, noise traders coming to the market with large orders, informed traders taking advantage of their superior or privileged information, and/or because market orders arrive at times when there is not enough depth at the best quotes in the LOB to fully absorb the orders.

We illustrate the behavior and performance of the strategies by showing how an HF trader, who maximizes expected terminal wealth and faces capital and inventory constraints, trades off expected profits against the variance and distribution of: terminal Profit and Loss (PnL), terminal inventory, and lifetime of inventories. Our analysis shows that final inventory penalties (which proxies the cost of unwinding positions at expiry of the strategy) affect the distribution of inventories at the expiry of the strategy, but has a limited effect on the distribution of the PnL of the strategy. On the other hand, our results show that penalizing inventories throughout the entire strategy (the penalization affects inventory management but does not alter the PnL directly) has an important effect on the three risk metrics we propose and provides a useful tool to trade off expected profits against risks. For example, we show that there are ranges of the penalization parameter where a small increase in the parameter causes a considerable reduction in the risks borne by the HF trader (as given by our risk metrics), whilst expected profits are hardly affected.

Our results also show that during times when markets become one-sided, i.e. there are more sell than buy market orders (or vice versa), the HF trader adjusts her strategy to take advantage of the market momentum. For example, if it is expected that the price of the asset will drift upwards, the optimal strategy consists in quickly building a long position and then proceed to post limit buy and sell orders to make profits on the realized spread. The optimal postings are such that inventories revert to an optimal

time-dependent position which is positive most of the time, but very close to (and at) expiry the optimal inventory level becomes negative.

The rest of this paper is organized as follows. Section 2 formulates and solves the HF's optimization problem. Section 3 illustrates the behavior of the optimal limit orders posted by the HF trader as a function of the model parameters and inventories. Section 4 proposes risk metrics for HF trading strategies. Section 5 shows how the HF trader employs the risk metrics to depict the tradeoff between expected profits and the exposure to risks and it also discusses how the HF trader fine tunes her strategies to maximize expected profits whilst obeying risk controls. Section 6 discusses further properties of the optimal strategy. Finally, Section 7 concludes and in the appendix we collect proofs of our results.

2. The High Frequency Trader's Optimization Problem

2.1. Formulation of the High Frequency market making problem

The HF strategy we develop here is based on posting limit buy and sell orders to profit from the realized spread which is the expected gain from a roundtrip trade. The HF trader is continuously updating her positions in the LOB to reflect changes in the midprice of the traded asset as well as other state variables, such as inventories and shape of the LOB, that we discuss below. The midprice of the asset is the average of the best bid and best ask and we assume that market participants that populate the LOB are constantly updating their quotes as a result of new information arriving in the market. We assume that the midprice (or fundamental price) S_t of the traded asset satisfies

$$dS_t = \sigma dW_t + \int_{\mathbb{R}_+^2} y_1 [\mu^+(d\mathbf{y}, dt) - \mu^-(d\mathbf{y}, dt)] , \quad (1)$$

where $(W_t)_{0 \leq t \leq T}$ is a standard Brownian motion, σ is a positive constant, and $\mu^\pm(d\mathbf{y}, dt)$ are Poisson random measures² on $\mathbb{R}_+^2 \times [0, T]$, independent of one another and the Brownian motion $(W_t)_{0 \leq t \leq T}$, with mean measures $\nu^\pm(d\mathbf{y}, dt) = \lambda^\pm F^\pm(dy_1) G^\pm(dy_2) dt$ where the distribution $F^\pm$ satisfies $\int_0^\infty F(dy_1) = 1$, $\int_0^\infty y_1^2 F(dy_1) < +\infty$ (i.e., with finite second moments) and $G^\pm(dy_2) = \frac{1}{\kappa^\pm} e^{-\kappa^\pm y_2} dy_2$ with $\kappa^\pm > 0$. The distributions $F^\pm$ determine the size of the impact of trades on the midprice, while the distributions $G^\pm$ are related to the probability of limit orders at price y being filled. Below we discuss the role of $G^\pm$ further. The counting processes corresponding to the arrival of sell and buy market orders are $M_t^- = \int_0^t \int_{\mathbb{R}_+^2} \mu^-(d\mathbf{y}, ds)$ and $M_t^+ = \int_0^t \int_{\mathbb{R}_+^2} \mu^+(d\mathbf{y}, ds)$, respectively, and are therefore Poisson processes with intensities λ^- and λ^+ . We often find it convenient to work with the random variables $\epsilon^\pm$ whose distribution functions are $F^\pm$. As usual, we work on a completed filtered probability space $(\Omega, \mathbb{F}, \mathcal{F} = \{\mathcal{F}_t\}_{0 \leq t \leq T}, \mathbb{P})$ where $\mathcal{F}$ is the natural filtration generated by the midprice $(S_t)_{0 \leq t \leq T}$ and the counting processes $(N_t^\pm)_{0 \leq t \leq T}$ which count the number of filled sell/buy limit orders that the trader posts. We define $N^\pm$ more precisely below.³

²See Kühn and Stroh (2012) for a thorough development of Poisson random measures for limit order book modeling.

³For stock price models with HF data where the interarrival times of trades are not exponential see Cartea and Meyer-Brandis (2010) and Cartea (2012).

Intuitively, we can view the dynamics of the midprice as the sum of two components. The first component, the Brownian motion on the right-hand side of (1), captures the changes in the midprice that are due to information flows that reach all or some market participants who subsequently update their quotes. The other component, the jump process with increments $\epsilon^+ dM_t^+ - \epsilon^- dM_t^-$, represents the changes in the midprice caused by the arrival of market orders that have a permanent price impact. Market orders may come at times when there is enough liquidity in the market – hence prices remain unchanged or change by a negligible amount; they may arrive at times when liquidity is thin or the orders sent by traders with superior information and these trades have a permanent impact on prices. The impact of trading on the midprice may also be viewed as the action of informed traders. If an informed trader purchases (sells) shares, they will only do so if the asset price is known to be going up (down). The resulting increase (decrease) of the midprice following informed trading can be approximated by an immediate, and permanent, price impact as we model here. Note we also assume that the size of the jump in the mid-price at the time of a market-order is stochastically independent of the depth at which the market order eats into the book.

In practice market orders also have a temporary price impact which depends on the type of information that triggered the trade and the resiliency of the LOB. For example, if a large market order (or a series of one-side market orders) arrives it will eat into one side of the LOB and it will take some time before other market participants update their quotes. If the market order is a pure liquidity trade which is not motivated by private information about the fundamental value of the asset, the price impact is temporary and, everything else equal, the midquote will revert to the value prior to the arrival of the market order. The time it takes for the midquote to mean-revert is determined by how long it takes market participants to process market information (i.e. find out that the trade is purely a liquidity trade) and repost resting orders to replenish the book. In our model, market orders have either a permanent price impact or no price impact, but it is possible to extend it so that market orders have a transitory price impact.

In our setting HF traders maximize expected terminal wealth and are constrained by the amount of inventories they are willing to hold. As mentioned above, HF traders' main competitive advantage is speed. HF strategies that profit from the realized spread are designed to enter and exit the market over very short-term horizons (seconds, or at most minutes) because the shorter the time scales the more likely are HF traders to take advantage of speed to process information and execute roundtrip trades at a profit. In our framework speed comes into play in two ways. One, we assume that the HF trader is able to process information and send instructions to the exchange in microseconds. Speed enables her to place optimal buy and sell limit orders in the LOB, i.e. how far away from the midquote S_t is it optimal to post a buy limit order $S_{t-} - \delta_t^-$ and a sell limit order $S_{t-} + \delta_t^+$ (with $\delta_t^\pm > 0$), before other slower traders can make decisions or reach the exchange. And two, because HF traders exploit their speed advantage to open and close positions, optimal strategies generate a high turnover of inventories. Everything else equal, HF traders prefer strategies with very short expected holding periods because they are constrained by the amount of inventories they can hold long or short.

Thus, in the optimization problem that the HF trader solves, inventories q_t are penalized and managed as follows. First, inventories receive a penalty throughout the entire path of the strategy, and, at maturity final inventory q_T is liquidated using market orders which are executed at worse prices than the best available quotes. The use of market orders is costly because there is an exchange fee for taking liquidity as

well as an adverse impact on prices which is increasing, and generally convex, in the size of the inventories; a market order will eat into the LOB away from the best quote. Finally, to avoid accumulating large positions, the HF trader imposes a cap on the number of shares that she can hold long or short at any point in time.

To formalize the investor's problem we need to introduce some more notation. Let N_t^- and N_t^+ denote the counting processes for the market orders that filled the HF trader's buy and sell limit orders. Thus, the total inventory of the HF trader is given by $q_t = N_t^- - N_t^+$ and so $q_t \in \mathbb{Z}$ for each t . Upon a buy or sell order being filled, the HF trader pays $(S_{t-} - \delta_t^-)$ or receives $(S_{t-} + \delta_t^+)$. We assume that a limit order at price level $S_{t-} \pm \delta_t^\pm$ is filled (given that a market order arrives) with probability $e^{-\kappa^\pm \delta_t^\pm}$. Hence, $\kappa^\pm$ can be interpreted as the exponential decay factor for the fill rate of orders placed away from the midprice, and the fill rate function can be written as $\Lambda_t^\pm = \lambda^\pm e^{-\kappa^\pm \delta_t^\pm}$ which is the rate of execution of a limit order. For queuing models of the LOB we refer the reader to Cont et al. (2010). For simplicity we assume that the LOB is instantly replenished after a market order arrives and that the spread is zero.⁴

In terms of the Poisson random measures $\mu^\pm(d\mathbf{y}, dt)$ introduced earlier, the counting processes corresponding to the trader's filled limit orders are written as $N_t^\pm = \int_0^t \int_{\mathbb{R}_+ \times [\delta_s^\pm, +\infty)} \mu^\pm(d\mathbf{y}, ds)$. Note that $N_t^\pm$ counts those events which lie above $y_2 > \delta_t^\pm$ (representing market orders large enough to eat into that level of the limit order book) regardless of the impact size y_1 on the mid-price. Recall that the filtration $\mathcal{F}$ is the natural one generated by both $(S_t)_{0 \leq t \leq T}$ and $(N_t^\pm)_{0 \leq t \leq T}$ and the jumps of S_t coincide with those of $M_t^\pm$, therefore the collection of compensated counting processes $(\widehat{N}_t^\pm, \widehat{M}_t^\pm)_{0 \leq t \leq T}$ where $\widehat{N}_t^\pm = N_t^\pm - \int_0^t \lambda^\pm e^{-\kappa^\pm \delta_s^\pm} ds$ and $\widehat{M}_t^\pm = M_t^\pm - \lambda^\pm t$ are $\mathcal{F}$ -martingales.

Finally, the HF trader's wealth X_t satisfies the SDE

$$dX_t = (S_{t-} + \delta_t^+) dN_t^+ - (S_{t-} - \delta_t^-) dN_t^-, \quad (2)$$

and the investor seeks the strategy $(\delta_s^\pm)_{t \leq s \leq T}$ which maximizes expected terminal wealth whilst penalizing and constraining inventories. Therefore, the HF trader solves the control problem

$$H(t, x, q, S) = \sup_{(\delta_s^\pm)_{t \leq s \leq T} \in \mathcal{A}} \mathbb{E} \left[X_T + q_T(S_T - \alpha q_T) - \phi \sigma^2 \int_t^T q_s^2 ds \mid X_{t-} = x, q_{t-} = q, S_{t-} = S \right], \quad (3)$$

where $q_t \in \mathbb{Z}$ is the inventory position at time t , the terminal date of the strategy is $T > t$, and $\mathcal{A}$ denotes the set of admissible strategies – strategies that are $\mathcal{F}_t$ -predictable processes such that inventories are bounded above by $\bar{q} > 0$ and below by $\underline{q} < 0$ both finite. Here, $\phi \geq 0$ penalizes deviations of q_t from zero along the entire path of the strategy, while the term $-\alpha q_T^2$ only penalizes deviations away from zero at maturity itself.⁵ Moreover, admissible strategies must obey upper and lower bounds on inventories:

⁴Our model could be extended so that the shape and depth of the book undergoes temporary changes in response to the arrival of market orders that eat into the book and to reflect the update of quotes due to the arrival of news. Moreover, a constant non-zero bid-ask spread can be incorporated, however, the qualitative results do not change and essentially result in shifting the optimal drift by the half-spread.

⁵In the work of Guéant et al. (2011) final inventory is penalized with $-\alpha|q_T|$. In principle we can include any liquidation penalty as a function of q or any inventory integrated penalty that is an even function of q .

$\underline{q} \leq q_s \leq \bar{q}, t \leq s \leq T$, where $\underline{q}, \bar{q} \in \mathbb{Z}$ with $-\infty < \underline{q} < 0 < \bar{q} < +\infty$. In particular, if $q_t = \bar{q}$, then the investor places only limit sell orders (i.e., she sets $\delta^- = +\infty$) and if $q_t = \underline{q}$, then the investor places only limit buy orders (i.e., she sets $\delta^+ = +\infty$).

The penalty term $\phi \sigma^2 \mathbb{E} \left[\int_t^T q_s^2 ds | \mathcal{F}_t \right]$ can further be motivated by noting that the total value of the investor's inventory (assuming a mark-to-market value at midprice and ignoring jumps) is $V = \int_t^T q_s dS_t^c$, where S^c denotes the continuous component of the midprice S . Hence its variance is

$$\mathbb{E} \left[\left(\int_t^T q_s dS_s^c \right)^2 \middle| \mathcal{F}_t \right] = \mathbb{E} \left[\int_t^T q_s^2 d[S^c, S^c]_s \middle| \mathcal{F}_t \right] = \sigma^2 \mathbb{E} \left[\int_t^T q_s^2 ds \middle| \mathcal{F}_t \right]$$

which, up to the scale factor ϕ , is the penalty we employ. In particular $\phi \geq 0$ reflects how severe the inventory controls are.

2.2. The Feedback Control of the Optimal High Frequency Trading Strategy

To solve the optimal control problem described above, a dynamic programming principle holds and the value function (3) is the unique viscosity solution of the HJB equation (see, e.g., Fleming and Soner (2006), Chap V):

$$\begin{aligned} 0 = & \partial_t H + \frac{1}{2} \sigma^2 \partial_{SS} H - \phi \sigma^2 q^2 \\ & + \lambda^+ \sup_{\delta^+} \left\{ e^{-\kappa^+ \delta^+} \mathbb{E} (H(t, x + (S + \delta^+), q - 1, S + \epsilon^+) - H) \right. \\ & \quad \left. + (1 - e^{-\kappa^+ \delta^+}) \mathbb{E} (H(t, x, q, S + \epsilon^+) - H) \right\} \\ & + \lambda^- \sup_{\delta^-} \left\{ e^{-\kappa^- \delta^-} \mathbb{E} (H(t, x - (S - \delta^-), q + 1, S - \epsilon^-) - H) \right. \\ & \quad \left. + (1 - e^{-\kappa^- \delta^-}) \mathbb{E} (H(t, x, q, S - \epsilon^-) - H) \right\}, \end{aligned} \quad (4)$$

subject to the terminal condition

$$H(T, x, q, S) = x + q(S - \alpha q), \quad (5)$$

and where x , S and q are the quantities at $t-$ (and not t) and the expectation is over the random variables $\epsilon^\pm$ and not over x , S , or q . Moreover, recall that the set of admissible strategies imposes bounds on q_t , this means that when $q_t = \bar{q}$ ($\underline{q}$) the optimal strategy is to post one-sided limit orders which are obtained by solving (4) with the term proportional to λ^- (λ^+) absent. Alternatively, one can view these boundary cases as imposing $\delta^- = +\infty$ ($\delta^+ = +\infty$) when $q = \bar{q}$ ($\underline{q}$).

Intuitively, the various terms in the HJB equation represent the arrival of market orders that may be filled by limit orders together with the diffusion of the asset price through the term $\frac{1}{2} \sigma^2 \partial_{SS} H$ and the effect of penalizing deviations of inventories from zero along the entire path of the strategy which is captured by the term $\phi \sigma^2 q^2$. The sup over δ^+ contain the terms due to the arrival of a market buy order (which is filled by a limit sell order). The first term represents the expected change in the value function H due

to the arrival of a market order which fills the limit order and the midquote S_t jumps up by the random amount ϵ^+ ; and the second term represents the arrival of a market order which does not reach the limit order's price level (but still causes a random jump in the midquote). Similarly, the sup over δ^- contain the analogous terms for the market sell orders which are filled by limit buy orders.

To solve the HJB equation we use the terminal condition (5) to make an ansatz for H . In particular, write

$$H(t, x, q, S) = x + q(S - \alpha q) + h_q(t), \quad (6)$$

which leads to a coupled system of ODEs for the functions $h_q(t)$ and allows us to solve for the optimal feedback controls. Before proceeding, note that the function $h_q(t)$ is related to the so called *reservation price* of the asset as in Ho and Stoll (1981). In particular, the reservation bid price r_t^- satisfies $H(t, x - r_t^-, q + 1, s) = H(t, x, q, s)$ and the reservation ask price r_t^+ satisfies $H(t, x + r_t^+, q - 1, s) = H(t, x, q, s)$. In other words, the reservation bid (ask) price is the price at which the HF trader is indifferent between buying (selling) and the inventory increasing (decreasing) by one unit or not trading at all.

Let, $\epsilon^\pm = \mathbb{E}[\epsilon^\pm]$ be the mean jump size in asset price when a market-order arrives. Then, substituting (6) in the HJB equation results in $h_q(t)$ satisfying

$$\phi \sigma^2 q^2 = \begin{cases} \partial_t h_q + \lambda^+ \sup_{\delta^+} \left\{ \epsilon^+ q + e^{-\kappa^+ \delta^+} (\delta^+ - \epsilon^+ - \alpha(1 - 2q) + h_{q-1} - h_q) \right\} \\ \quad + \lambda^- \sup_{\delta^-} \left\{ -\epsilon^- q + e^{-\kappa^- \delta^-} (\delta^- - \epsilon^- - \alpha(1 + 2q) + h_{q+1} - h_q) \right\}, & q \neq \underline{q}, \bar{q}, \\ \partial_t h_q + \lambda^+ \sup_{\delta^+} \left\{ \epsilon^+ q + e^{-\kappa^+ \delta^+} (\delta^+ - \epsilon^+ - \alpha(1 - 2q) + h_{q-1} - h_q) \right\} - \epsilon^- \lambda^- q, & q = \bar{q}, \\ \partial_t h_q + \lambda^- \sup_{\delta^-} \left\{ -\epsilon^- q + e^{-\kappa^- \delta^-} (\delta^- - \epsilon^- - \alpha(1 + 2q) + h_{q+1} - h_q) \right\} + \epsilon^+ \lambda^+ q, & q = \underline{q}, \end{cases} \quad (7)$$

subject to $h_q(T) = 0$ which allows us to solve for the optimal feedback controls, in terms of $h_q(t)$, as shown in the proposition below.

Proposition 1. Optimal Feedback Controls. *The optimal feedback controls of the HJB equation are given by*

$$\delta^{+*}(t, q) = \frac{1}{\kappa^+} + \epsilon^+ + \alpha(1 - 2q) - h_{q-1}(t) + h_q(t), \quad q \neq \underline{q}, \quad \text{and} \quad (8a)$$

$$\delta^{-*}(t, q) = \frac{1}{\kappa^-} + \epsilon^- + \alpha(1 + 2q) - h_{q+1}(t) + h_q(t), \quad q \neq \bar{q}. \quad (8b)$$

PROOF. Applying the first order conditions directly to each sup term in the HJB equation (7) leads immediately to the result. $\square$

To understand the intuition behind the feedback controls we first note that the optimal $\delta^\pm$ can be decomposed into four terms. The first component, $1/\kappa^\pm$, is the optimal strategy that a risk-neutral agent would employ in the absence of both terminal date restrictions (i.e. $T = \infty$, below in Proposition 6 we study the limiting case when $T \rightarrow \infty$) and inventory constraints. To see this, note that the expected gains from buying (selling) the asset at the midprice S_t , followed by selling (buying) it using a limit order at $S_{t-} + \delta^+$ ($S_{t-} - \delta^-$), is given by $\delta^\pm e^{-\kappa^\pm \delta^\pm}$, which is maximized if the limit order is posted $1/\kappa^\pm$ away from the midprice.

The second term shows that the HF trader incorporates the expectation of the jump in prices, conditioned on a market order arriving, by posting limit orders which are $\varepsilon^\pm = \mathbb{E}[\epsilon^\pm]$ further away from the midprice. In this way the HF trader recovers, on average, the losses she incurs when her limit orders are filled by trades that have a permanent price impact – for example, losses arising from trading with more informed traders (adverse selection).

Furthermore, the term $\alpha(1 \pm 2q)$ ensures that positions neither become too large nor too small, relative to the optimal inventory level, because at time T there are liquidation costs. For example, the time T optimal spreads are given by $\delta^\pm = 1/\kappa^\pm + \varepsilon^\pm + \alpha(1 \mp 2q)$ and one can see that when $q > 0$ ($q < 0$), the limit sell (buy) order is posted nearer to the midprice to increase the likelihood of selling (buying) the stock, which induces inventories to revert to zero.

Finally, the term $-h_{q-1}(t) + h_q(t)$, controls for inventories through time. As expected, if inventories are long, then the strategy consists in posting limit orders that increase the probability of sell orders being hit. Moreover, the function $h_q(t)$ also induces mean reversion to an optimal inventory level as a result of penalizing accumulated inventories throughout the entire trading horizon and the strategy approaching T as well as the other parameters of the model, including $\phi\sigma^2$.

Proposition 2. Solving Reduced Value Function. *If $\kappa^+ = \kappa^- = \kappa$, then write $h_q(t) = \frac{1}{\kappa} \ln \omega_q(t)$ and stack $\omega_q(t)$ into a vector $\boldsymbol{\omega}(t) = [\omega_{\bar{q}}(t), \omega_{\bar{q}-1}(t), \dots, \omega_{\underline{q}}(t)]'$. Further, let $\mathbf{A}$ denote the $(\bar{q} - \underline{q} + 1)$ -square matrix whose rows are labeled from $\bar{q}$ to $\underline{q}$ and whose entries are given by*

$$\mathbf{A}_{i,q} = \begin{cases} q \kappa (\varepsilon^+ \lambda^+ - \varepsilon^- \lambda^-) - \phi \sigma^2 \kappa q^2, & i = q, \\ \lambda^+ e^{-\kappa(\frac{1}{\kappa} + \varepsilon^+ + \alpha(1-2q))}, & i = q - 1, \\ \lambda^- e^{-\kappa(\frac{1}{\kappa} + \varepsilon^- + \alpha(1+2q))}, & i = q + 1, \\ 0, & \text{otherwise.} \end{cases} \quad (9)$$

Then,

$$\boldsymbol{\omega}(t) = e^{\mathbf{A}(T-t)} \mathbf{1}, \quad (10)$$

where $\mathbf{1} = [1, \dots, 1]'$ is a $(\bar{q} - \underline{q} + 1)$ -dim vector of ones.

PROOF. See Appendix A.1.

As a direct consequence of Proposition 2, when the market is symmetric the spread on the buy side with q shares equals the spread on the sell side with $-q$ shares.

Corollary 3. Symmetry in Spreads. *When $\kappa^\pm = \kappa$, $\lambda^\pm = \lambda$, $\varepsilon^\pm = \varepsilon$, the optimal spreads are symmetric in the sense that $\delta^{\pm*}(t, q) = \delta^{\mp*}(t, -q)$.*

PROOF. See Appendix A.2.

Since we now have a classical solution of the HJB equation (7) which is also bounded above and below, it is easy to see that the feedback control in (8) are admissible and the verification theorem below follows.

Theorem 4. Verification. *Let $h_q(t)$ be a classical solution to (7). Then $H(t, w, S, q) = x + q(S - \alpha q) + h_q(t)$ is the value function of the trader's control problem (3) and the optimal controls are given by (8) in feedback form.*

PROOF. See Appendix B.

In the following proposition we show that the optimal HF strategy $\delta^{*\pm}(t, q)$ induces mean reversion in inventories – below in the simulations section we show that the speed of mean reversion is increasing in the penalty parameter ϕ .

Corollary 5. Mean Reversion in Inventories. *Given the pair of optimal strategies $\delta^+(t, q), \delta^-(t, q)$, the expected drift in inventories q_t is given by*

$$\mu_t(q) \triangleq \lim_{s \downarrow t} \frac{1}{s - t} \mathbb{E}[q_s - q_t | q_t = q] = \lambda^- e^{-\kappa^- \delta^-(t, q)} - \lambda^+ e^{-\kappa^+ \delta^+(t, q)}. \quad (11)$$

PROOF. It is straightforward to see that the expected drift in inventories is given by the difference in the arrival rates of filled orders. $\square$

There are two remaining interesting limiting cases to study. The first is in the limit of being “far” from the investor's maturity and the second is in the limit of being “near” to the investor's maturity. Since the matrix $\mathbf{A}$ has elements which are linear in the activity rates, the time scale on which the “far” and “near” limits are valid is of the order $\min(1/\lambda^+, 1/\lambda^-)$. The key mathematical results are recorded below.

Proposition 6. Limiting Spreads. Assume that $\kappa^\pm = \kappa$ and let $\delta_0^\pm(q) = \frac{1}{\kappa} + \varepsilon^\pm + \alpha(1 \mp 2q)$. Then,

1. As $T \rightarrow \infty$, the limiting spreads are given by the (time independent) values

$$\delta^+(t, q) = \delta_0^+(q) - c^+(q), \quad q \neq \underline{q}, \quad \text{and} \quad \delta^-(t, q) = \delta_0^-(q) - c^-(q), \quad q \neq \bar{q}, \quad (12)$$

where $c^\pm(q) = \mathbf{v}_{q \mp 1} - \mathbf{v}_q$ and the vector $\mathbf{v} = \frac{1}{\kappa} \ln(\mathbf{U}[\mathbf{U}^{-1}\mathbf{1}]_\ell)$ where the logarithm is taken componentwise, ℓ denotes the index of the largest eigenvalue of the matrix $\mathbf{A}$, the notation $[\cdot]_\ell$ denotes the vector obtained by setting all entries to zero except for the ℓ^{th} entry, and $\mathbf{U}$ denotes the matrix whose columns are the eigenvectors of $\mathbf{A}$.

2. As $t \uparrow T$, the limiting spreads for $q \neq \bar{q}, \underline{q}$ are given by

$$\begin{aligned} \delta^+(t, q) = \delta_0^+(q) + & \left[\frac{2}{\kappa} \left(\lambda^+ e^{-\kappa(\frac{1}{\kappa} + \varepsilon^+ - 2\alpha q)} - \lambda^- e^{-\kappa(\frac{1}{\kappa} + \varepsilon^- + 2\alpha(q-1))} \right) \sinh(\kappa\alpha) \right. \\ & \left. + (\lambda^+ \varepsilon^+ - \lambda^- \varepsilon^-) - \phi \sigma^2 \kappa(2q-1) \right] (T-t) + o(T-t), \end{aligned} \quad (13a)$$

$$\begin{aligned} \delta^-(t, q) = \delta_0^-(q) + & \left[\frac{2}{\kappa} \left(-\lambda^+ e^{-\kappa(\frac{1}{\kappa} + \varepsilon^+ - 2\alpha(q+1))} + \lambda^- e^{-\kappa(\frac{1}{\kappa} + \varepsilon^- + 2\alpha q)} \right) \sinh(\kappa\alpha) \right. \\ & \left. - (\lambda^+ \varepsilon^+ - \lambda^- \varepsilon^-) + \phi \sigma^2 \kappa(2q+1) \right] (T-t) + o(T-t). \end{aligned} \quad (13b)$$

PROOF. See Appendix A.3.

As direct consequence of the previous result we have the following limiting behavior of the strategies.

Corollary 7. Terminal Strategy Approach. When $\kappa^\pm = \kappa$, $\lambda^\pm = \lambda$, $\varepsilon^\pm = \varepsilon$ and $\phi = 0$, we have for $q \neq \bar{q}, \underline{q}$

$$0 > \begin{cases} \lim_{t \uparrow T} \partial_t \delta^+(t, q), & \forall q > 0, \\ -\lim_{t \uparrow T} \partial_t \delta^+(t, q), & \forall q \leq 0, \end{cases} \quad \text{and} \quad 0 < \begin{cases} \lim_{t \uparrow T} \partial_t \delta^-(t, q), & \forall q \leq 0, \\ -\lim_{t \uparrow T} \partial_t \delta^-(t, q), & \forall q < 0. \end{cases} \quad (14)$$

This result tells us whether the strategy approaches its terminal strategy of $\delta_0^\pm(q)$ from above or below. For example, if $q \leq 0$ then $\delta^+(t, q)$ approaches $\delta_0^+(q)$ from below, while if $q > 0$, then $\delta^+(t, q)$ approaches $\delta_0^+(q)$ from above.

3. The Behavior of the Strategy

In this section we illustrate several aspects of the behavior of the optimal strategy as a function of $q, \phi, \alpha, \lambda^\pm, \bar{q}, -\underline{q}$. In all the examples of this section the terminal date of the strategy is $T = 10$ seconds,

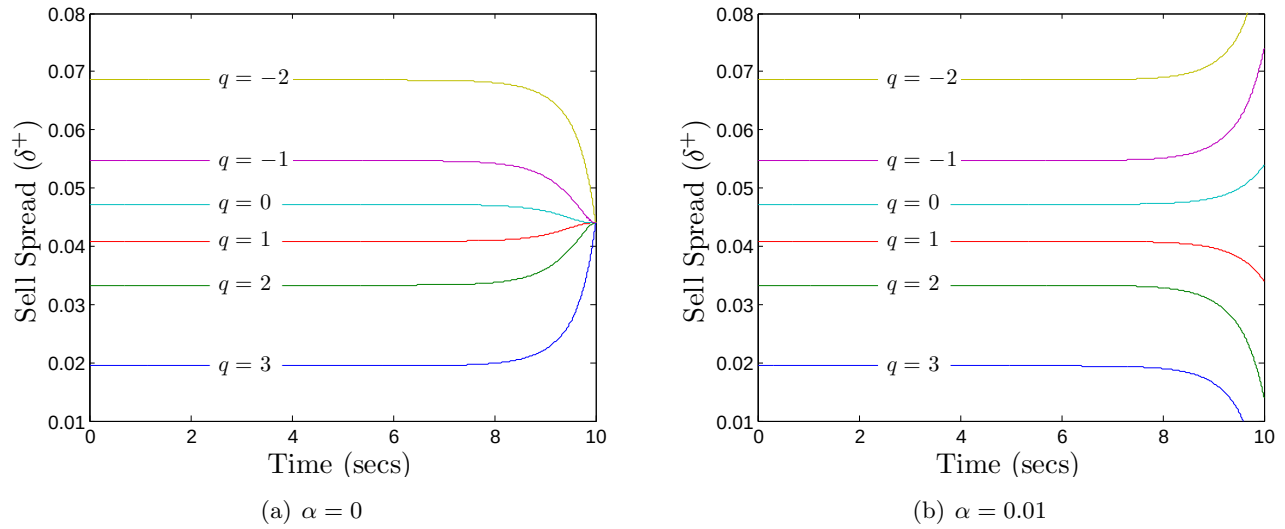


Figure 1: The optimal spreads as a function of time for various inventory levels and $T = 10$. The remaining model parameters are: $\lambda^\pm = 2$, $\kappa^\pm = 25$, $\varepsilon^\pm = \frac{1}{250}$, $\bar{q} = -\underline{q} = 3$, $\phi = 0$, $\sigma = 0.1$. Only the sell-side is shown as the buy-side is similar.

$\kappa^\pm = 25$, $\sigma = 0.1$, and upon the arrival of market orders, the midprice jumps up or down by the amount $1/25$, with probability $1/10$, or 0 , with probability $9/10$, so that $\mathbb{E}[\epsilon^\pm] = \varepsilon^\pm = 1/250$.

3.1. Liquidation penalty and the optimal spreads as a function of time for various inventory levels

Figure 1 shows the behavior of the optimal spreads as a function of the liquidation penalty $\alpha = \{0, 0.01\}$ for different inventory levels. In the examples the arrival rate of market orders is $\lambda^\pm = 2$ (there are on average 2 buy and 2 sell market orders per second), $\bar{q} = -\underline{q} = 3$, and $\phi = 0$. In this subsection we discuss the results for the sell spread, the buy spread has a similar interpretation. In panel (a) the HF strategy is not penalized for unwinding inventories at expiry, $\alpha = 0$, and we show the optimal sell postings δ^+ , i.e. upon the arrival of a market buy order the HF trader is willing to sell one unit of the asset at the price $S_t + \delta^+$. Thus, when $\alpha = 0$ and inventories are close to the allowed minimum, the optimal posting is furthest away from the midquote because only at a very ‘high’ price is the HF trader willing to decrease her inventories further. However, as the strategy approaches T and $q_t < 0$, the optimal spread δ^+ decreases. To understand the intuition behind the optimal strategy note that if the terminal inventory $q_T < 0$ is liquidated at the price $S_T - \alpha q_T$, then when α is sufficiently low and $q_t < 0$, as well as being fractions of a second away from expiry, it is optimal to post nearer the midprice to increase the chances of being filled (i.e. selling one more unit of the asset) because the price is not expected to move too much before expiry and the entire position will be unwound at the midprice – making a profit on the last unit of the asset that was sold.

When $q_t > 0$ and the strategy is far away from T it is optimal to post low spreads because there is enough time before expiry to turn over positions. On the other hand, when $q_t \geq 0$ and the strategy approaches T , it is optimal to post further away from the midquote. Initially this result appears to be counterintuitive because one would expect that it would be best to post nearer the midprice, rather than further, to

increase the probability of selling the asset at a price slightly above S_t to reduce the long position instead of closing the position at S_T . Note however, that if the strategy is one second away from expiry, the HF trader knows that on average there will be 2 buy/sell market incoming orders and that there is a tradeoff between posting nearer or further away from the midprice. The tradeoff consists in comparing what is the change in expected wealth from posting a limit sell order far from the midprice compared to the change in the expected wealth from posting close to the midprice. The former has a lower probability of being filled than the latter, but overall the HF trader is better off, in expectation, if she posts the sell limit order deeper in the LOB as shown by panel (a) in the figure.

When the penalty for liquidating terminal inventory is increased, as in panel (b), we observe a change in the qualitative behavior of the strategy as T is approached. For instance, panel (b) shows that when $q_t < 0$ and the strategy approaches T , the spreads increase – if the HF trader sells another asset close to expiry, it is at a price that ensures that it would cover the liquidation penalty of αq_T per share. Similarly, when $q_t > 0$ and the strategy approaches T , it is optimal to post very close to the midprice to increase the probability of unwinding the long position with a limit order before T rather than picking up the liquidation penalty.

3.2. The optimal spread at zero inventory as a function of time for various levels of ϕ and α

Figure 2 shows how the optimal buy/sell spread changes as a function of the liquidation penalty parameter α and the inventory penalty ϕ when $q_t = 0$. We recall that due to the symmetry of both the arrival rate of market orders and the size of jumps in the midprice, the buy and sell spreads are the same when $q_t = 0$ (as shown in Corollary 3). In our discussion we focus on the sell spread, the interpretation of the buy spread is similar. In the left panel, $\alpha = 0$ and we see that when $t < T$ the higher is the inventory penalty ϕ , the further away from the midprice is the optimal sell spread, but as the strategy approaches T , all spreads, as a function of ϕ , converge to the same terminal value of $\frac{1}{\kappa^\pm} + \varepsilon^\pm$. The monotonic decreasing behavior is natural because, here, there is no terminal penalty and the running penalty (driven by ϕ) does not have time to constrain the strategy.

The picture on the right-hand shows how the liquidation penalty affects the optimal sell strategy. We observe that for low values of the liquidation parameter α the optimal spread starts to decrease a few seconds before T and then increases. The intuition behind this behavior is the following. When the strategy approaches expiry, and $q_t = 0$, it is optimal to lower the sell spread in the hope of acquiring a unit of the asset and have enough time left for other market orders to arrive on the other side of the LOB to be able to unwind the position. However, if the strategy is approaching T then there is not enough time left to execute a roundtrip trade and this explains why the spread turns and increases very close to T . This can be seen as a direct consequence of (i) Corollary 7 which shows that the strategy (for $q = 0$) must approach its terminal limit δ_0^+ from below, and (ii) the fact that the terminal limit lies below the far from maturity limit $\delta^+(t, q)$. On the other hand, for high values of α we see that as the strategy approaches expiry, the spread increases because if the offer is lifted by an incoming buy market order, the HF trader has very little time left to unwind her position before expiry and the spread has to be high enough to cover the high liquidation penalty.

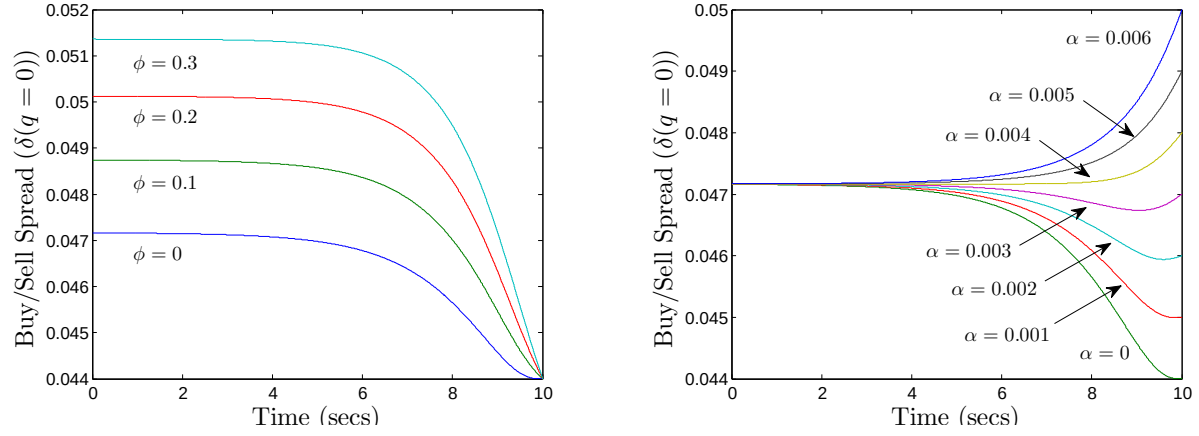


Figure 2: The optimal spread at $q = 0$ as a function of time for various levels of ϕ and α . The remaining model parameters are: $T = 10$, $\lambda^{\pm} = 2$, $\kappa^{\pm} = 25$, $\varepsilon^{\pm} = \frac{1}{250}$, $-q = \bar{q} = 3$, and $\sigma = 0.1$. In the left panel $\alpha = 0$ and in the right panel $\phi = 0$.

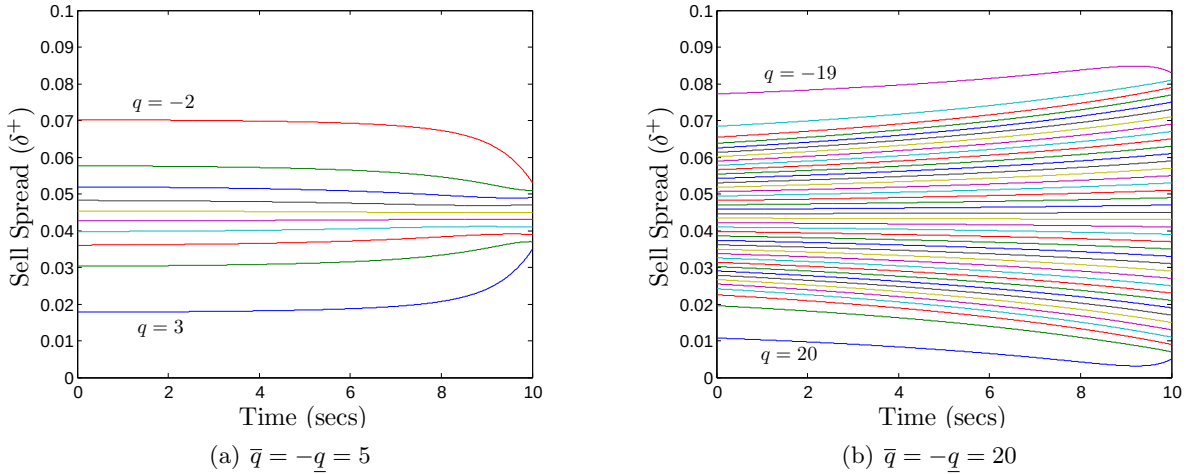


Figure 3: The optimal spreads as a function of time for two inventory levels and maximal positions with $T = 10$. The remaining model parameters are: $\lambda^{\pm} = 2$, $\kappa^{\pm} = 25$, $\varepsilon^{\pm} = \frac{1}{250}$, $\phi = 0$, $\sigma = 0.1$ and $\alpha = 0.001$.

3.3. The optimal spreads as a function of time for various inventory levels and maximal positions

Figure 3 shows how the optimal sell spread depends on the maximum amount of shares q (long or short) that the HF trader is allowed to hold. We assume that $T = 10$ seconds, $\lambda^{\pm} = 2$, $\kappa^{\pm} = 25$, $\varepsilon^{\pm} = \frac{1}{250}$, $\phi = 0$, $\sigma = 0.1$, and $\alpha = 0.001$. As expected, as inventories get closer to $\underline{q} + 1$, which are given by the lines with highest spread in each panel, the sell spread increases because only if the expected realized spread is higher will the HF trader be willing to increase her short position in the asset. The Figure also shows that when the strategy is approaching T the optimal spread decreases or increases when inventory constraints are $\underline{q} + 1$ or $\bar{q}$ respectively. Finally, it is interesting to see that as $\bar{q}$ ($\underline{q}$) increases (decreases), the maximum and minimum spread, which correspond to inventories reaching the largest position (long or short) that can be held, do not vary much when inventory bounds increase from 5 to 20. In fact, we observe that the effect of a larger q is to reduce the difference between optimal spreads that only differ by one unit of

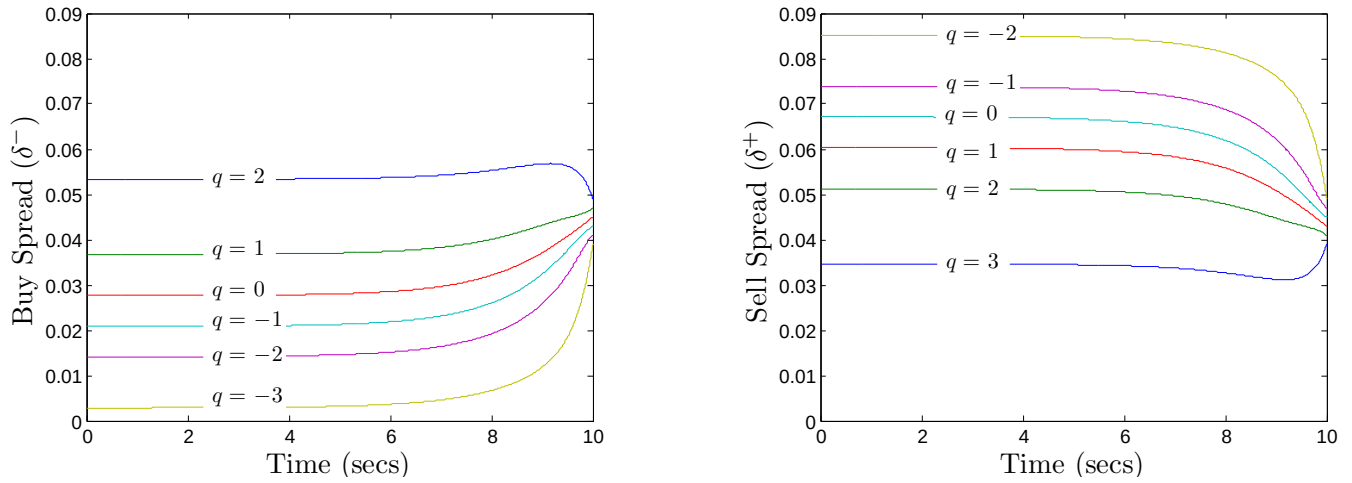


Figure 4: The optimal spreads as a function of time for various inventory levels when buy market orders are more likely $\lambda^+ = 6$, $\lambda^- = 4$. The remaining model parameters are: $T = 10$, $\kappa^\pm = 25$, $\varepsilon^\pm = \frac{1}{250}$, $\bar{q} = -\underline{q} = 3$, $\phi = 0$, $\sigma = 0.1$ and $\alpha = 0.001$.

inventory.

3.4. The optimal spreads as a function of time for various inventory levels when buy market orders are more likely

Figure 4 shows the behavior of the optimal buy and sell spread when the arrival of market orders is not symmetric. We assume that the intensity of arrival of buy market orders is higher than that of sell orders, $\lambda^+ = 6$, $\lambda^- = 4$, and that $T = 10$, $\kappa^\pm = 25$, $\varepsilon^\pm = \frac{1}{250}$, $\bar{q} = -\underline{q} = 3$, $\phi = 0$, $\sigma = 0.1$ and $\alpha = 0.001$. The figure shows that for all levels of inventory, the postings around the midprice will show a larger spread on the sell than on the buy side because the HF trading strategy incorporates the fact that the value of the asset is drifting upwards as a consequence of the arrival of more buy than sell market orders. Thus, when the strategy is not close to expiry, posting lower spreads on the buy side means that the HF trader expects to build a long inventory position because inventories will on average appreciate in value due to the upward pressure in prices. Finally, as the strategy approaches expiry, spreads on both sides of the book become more symmetric to ensure that inventories are unwound.

3.5. Mean reversion of inventories: speed and level

In Figure 5 we assume that the strategy is far away from expiry and we illustrate how the inventory drift (see Proposition 5) varies as a function of inventory and how this rate depends on the mean arrival rate of market orders and the parameter ϕ that penalizes deviations of inventories from zero. In the left panel, the arrival rates of market buy and sell orders are equal and we observe that the drift changes sign at zero inventory level (for positive inventories the drift is negative, while for negative inventories the drift is positive), thus the HF trader's inventory mean-reverts to zero. The figure also shows that the higher the penalty ϕ , the quicker will inventories mean revert to zero.

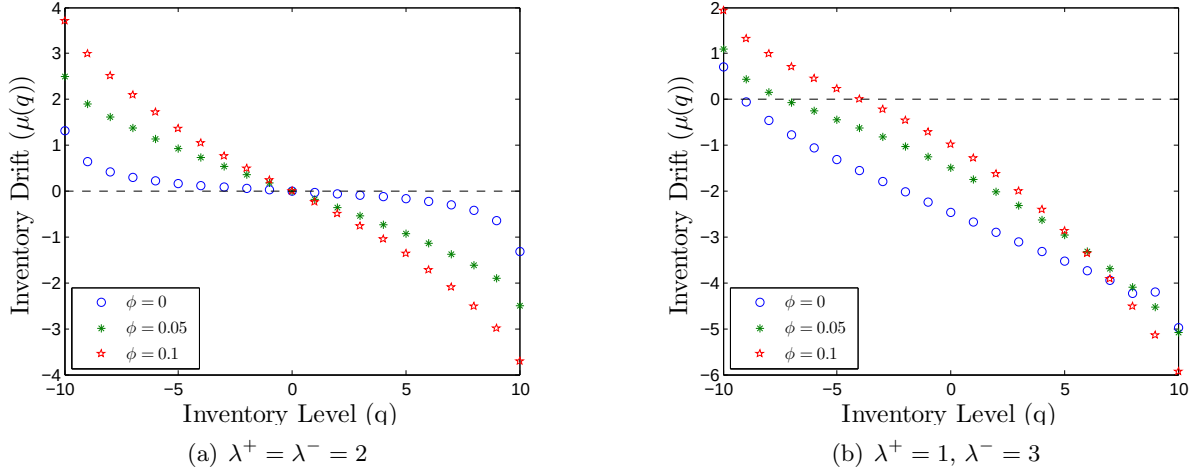


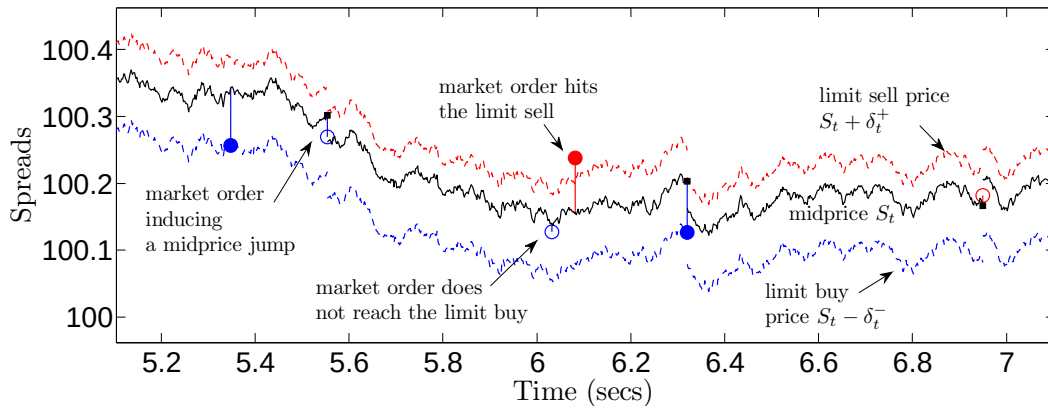
Figure 5: Inventory drift $\mu(q)$ for each level of inventory with equal and unequal arrival rate of market orders, $\bar{q} = -\underline{q} = 10$, $\alpha = 0$, $\varepsilon^\pm = \frac{1}{250}$, and $\kappa^\pm = 25$, and the strategy is far away from expiry.

In the right panel of Figure 5 the arrival rate of market sell orders is much larger than that of buy orders. We observe that the drift changes sign at negative inventory levels, and consequently, the HF trader's inventory mean reverts to holding a short position in the asset in anticipation of the downward pressure on prices due to the excess arrival of sell market orders. Recall that the midquote price process is given by (1) so the HF trader knows that the midprice is drifting downward, $\lim_{\Delta t \downarrow 0} \frac{1}{\Delta t} \mathbb{E}[S_{t+\Delta t} - S_t | \mathcal{F}_t] = -\frac{2}{250}$, in which case it is optimal to always hold short positions, when far away from expiry, because the value of the asset holdings is expected to decrease. Moreover, if the penalty ϕ is increased, inventories mean revert to values which are closer to zero.

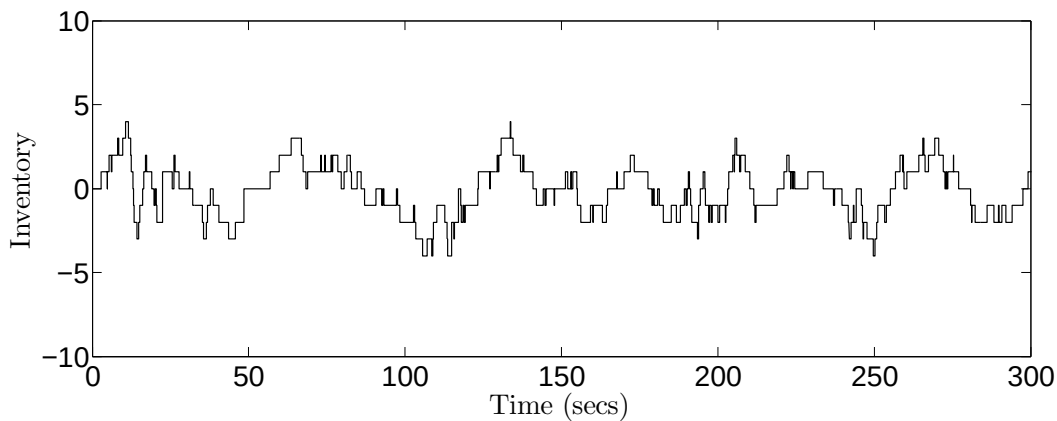
In panel (a) of Figure 6 we show a 2 second snippet of one simulation of the midprice where market buy and sell orders arrive with intensity $\lambda^\pm = 2$, and we also show how these market orders cross with the HF trader's limit orders. In panel (b) of the same figure we show the dynamics of the HF trader's inventory position over a 300 second window – the picture clearly shows that inventories are quickly mean reverting to zero and that they are always far away from the allowed upper and lower bound $\bar{q} = -\underline{q} = 10$.

4. Risk Metrics for High Frequency Trading Strategies

HF trading is relatively new and relies on strategies which, as any other type of investment in the stock market, seek to profit from buying and selling assets. However, one of the differences between HF trading and any other type of activities that we observe in the exchanges, is the speed at which HF traders are able to process information and make trading decisions. Moreover, because speed is at the core of their competitive advantage, it is logical that HF traders invest in skills to develop trading strategies based on completing roundtrip trades as quickly as possible which is one of the reasons that explain why inventories



(a) Midprice dynamics and optimal strategy over a 2 sec interval



(b) Inventory path over 300 sec interval

Figure 6: Strategy and inventory sample paths over a 2 sec and 300 sec interval respectively. The model parameters are $\kappa^\pm = 25$, $\lambda^\pm = 2$, $\epsilon^\pm = \frac{1}{25}\mathbf{a}^\pm$ where $\mathbf{a}^\pm$ are independent Bernoulli r.v. with success prob of $\frac{1}{10}$, $\sigma = 0.1$, $\bar{q} = -\underline{q} = 10$, $S_0 = 100$, $\phi = 0.1$ and $\alpha = 0.001$. Filled circles indicate market orders that hit/lift the limit order, unfilled circles indicate orders that did not reach the post limit order.

revert to an optimal level of inventory so rapidly.⁶

Therefore, everything else equal, HF traders prefer strategies which are less exposed to inventory risk. Holding larger positions than the optimal one is risky for HF traders, and holding smaller positions than the optimal one for a long period of time is also risky. Ideally, HF traders prefer to hold small positions (relative to what other types of market participants are holding) for a short period of time because the longer the holding period, the riskier it is for them because they lose their competitive advantage to the more traditional market participants – being able to take decisions in milliseconds becomes less effective and less valuable.

⁶This is another striking difference between the behavior of HF traders and other investors, such as traditional market makers, who are willing (and required by the exchange) to hold inventories for much longer periods of time.

Most risk metrics have been designed with the more traditional market participants in mind. For instance, the variance of the PnL is a widely used risk metric to assess an investment strategy that maximizes expected profits. However, this risk metric fails to address, in a direct rather than indirect way, the tradeoff between inventory risk and expected profits which is fundamental for HF trading. In the rest of this section we propose risk metrics for HF trading strategies and illustrate them using the model we developed above. We point out that the risk metrics proposed here have been designed to be applied to all HF trading strategies, not only the ones discussed in this paper. The risk metrics provide a tool to understand and quantify exposure to inventory risk (which also proxies for capital constraints), as well as to enable HF traders to fine tune their strategies by trading off inventory risk against expected profits.

4.1. Risk metrics for High Frequency trading

In this section we discuss three risk metrics designed to assess the risk of HF strategies which profit from making the realized spread in the intraday market. Thus, the risk metrics are for HF market makers that meet all or some of the following: cannot or do not wish to hold inventories that deviate from the optimal level for a long period of time; employ strategies which turn over a vast amount of positions in the intraday market but do not want to hold large inventories; are capital constrained; operate in markets where liquidity varies throughout the trading day.

The first risk metric quantifies the variance of the PnL of the strategy, and the other two metrics quantify the exposure that HF trading strategies have to inventory risk. Below q_t^* and X_t^* denote the path of inventory and cash holdings respectively, followed by a general HF strategy – i.e. not limited to the one developed above. We let $\mathbb{E}_t[\cdot]$ and $\mathbb{V}_t[\cdot]$ denote the expectation and variance operators conditional on $X_{t-} = x$, $q_{t-} = q$ and $S_{t-} = S$.

1. Variance of terminal PnL:

$$\mathcal{R}^1(t, x, q, S) \triangleq \mathbb{V}_t [X_T^* + q_T^* (S_T - \alpha q_T^*)] . \quad (15)$$

2. Variance of terminal inventory:

$$\mathcal{R}^2(t, q) \triangleq \mathbb{V}_t [q_T^*] . \quad (16)$$

3. Variance of lifetime inventory:

$$\mathcal{R}^3(t, q) \triangleq \sum_{n \in \{\underline{q}, \dots, \bar{q}\}} (n - \langle q^* \rangle_{t,q})^2 \mathbb{Q}_n(t, q) , \quad (17)$$

where $\langle q^* \rangle_{t,q} = \sum_{n \in \{\underline{q}, \dots, \bar{q}\}} n \mathbb{Q}_n(t, q)$ and

$$\mathbb{Q}_n(t, q) \triangleq \mathbb{E}_t \left[\frac{1}{T-t} \int_t^T \mathbb{I}\{q_s^* = n\} ds \right] , \quad (18)$$

denotes the percentage of time (from t to T) that the strategy holds an inventory of n given that the strategy begins with q -shares. Moreover, note that $\mathbb{Q}_n(t, q)$ defines a distribution over the discrete set $\{\underline{q}, \dots, \bar{q}\}$ of possible inventory levels and that risk metric $\mathcal{R}^3(t, q)$ assesses the strategy over the entire time horizon and not just at maturity.

For the particular optimal HF strategy derived in this paper we can compute risk metrics $\mathcal{R}_1$, $\mathcal{R}_2$ and $\mathcal{R}_3$ in closed-form in terms of time ordered exponentials. It is possible to further obtain an approximation of the three risk metrics by setting the strategy to the limiting strategy of Proposition 6 where the spreads are not time dependent. As we saw in the previous section, for the particular parameters we have chosen to study, the effect of the terminal date T starts to show only a couple of seconds before expiry. In general the scale on which the strategy approaches its asymptotic value is very short, consequently, using the limiting strategy provides us with an excellent approximation to the strategy of Proposition 1. In the proposition below, we use the limiting strategy to calculate in closed-form risk metrics $\mathcal{R}_1$, $\mathcal{R}_2$ and $\mathcal{R}_3$. In Appendix A.4 we show the fully time dependent risk metrics.

Proposition 8. Risk Metrics. *Suppose the agent follows the strategy specified by the limiting optimal spreads⁷ $\delta_q^{\pm*}$ given by Proposition 6. Let $\mathbf{B}$ denote the matrix whose elements (labeled from $\bar{q}$ to q) are*

$$\mathbf{B}_{i,q} = \begin{cases} -\left(\lambda^- e^{-\kappa \delta_q^{-*}} + \lambda^+ e^{-\kappa \delta_q^{+*}}\right), & i = q, \\ \lambda^+ e^{-\kappa \delta_q^{+*}}, & i = q - 1, \\ \lambda^- e^{-\kappa \delta_q^{-*}}, & i = q + 1, \\ 0, & \text{otherwise,} \end{cases} \quad (19)$$

and let $\mathbf{V}$ denote the matrix of eigenvectors stacked columnwise in decreasing order of its eigenvalues. Then we have the following:

1. $\mathbf{B}$ contains exactly one zero eigenvalue, and all other eigenvalues are negative.
2. The terminal variance risk metric is given by

$$\begin{aligned} \mathcal{R}^1(t, x, q) = & \mathbf{a}(t, q) - \mathbf{m}_1(t, q)^2 + 2\alpha q^2 \mathbf{m}_1(t, q) + (\mathbf{b}(q) - 2\mathbf{m}_1(t, q))x \\ & + (\mathbf{c}(t, q) - 2q\mathbf{m}_1(t, q))S \end{aligned} \quad (20)$$

where $\mathbf{m}_1(t, q)$ is provided in Equation (A.5), $\mathbf{a} = \mathbf{V}\tilde{\mathbf{a}}$, $\mathbf{b} = \mathbf{V}\tilde{\mathbf{b}}$, and $\mathbf{c} = \mathbf{V}\tilde{\mathbf{c}}$ with $\tilde{\mathbf{a}}$, $\tilde{\mathbf{b}}$ and $\tilde{\mathbf{c}}$ provided in Equations (A.10a), (A.10b) and (A.10c). Moreover, we have (with $\tau = T - t$)

$$\tilde{\mathbf{a}}_j(t, q) = r_1 \tau^3 (1 + o(\tau)), \quad \tilde{\mathbf{b}}_j(t, q) = \tilde{\chi}_1 \tau (1 + o(\tau)), \quad \tilde{\mathbf{c}}_j(t, q) = \frac{1}{2} \tilde{\Psi}_{11} \tilde{\chi}_1 \tau^2 (1 + o(\tau)), \quad (21)$$

for $j = 1$ and equal zero otherwise. Here, the constants $\tilde{\chi}_1$ and $\tilde{\Psi}_{11}$ are given in (A.11a) and r_1 is given in (A.11h).

⁷Recall these are independent of time and depend only on q .

3. The limiting terminal inventory risk metric is given by

$$\lim_{T \rightarrow +\infty} \mathcal{R}^2(t, q) = (\mathbf{V} [\mathbf{V}^{-1} \boldsymbol{\beta}]_1)_q - \left((\mathbf{V} [\mathbf{V}^{-1} \boldsymbol{\gamma}]_1)_q \right)^2 \quad (22)$$

where $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$ are $(\bar{q} - \underline{q} + 1)$ -dim vectors with elements $\beta_q = q^2$ and $\gamma_q = q$, respectively.

4. For each inventory level, the limiting lifetime is given by

$$\lim_{T \rightarrow +\infty} \mathbb{Q}_n(t, q) = (\mathbf{V} [\mathbf{V}^{-1} \mathbf{1}_n]_1)_q, \quad (23)$$

where $\mathbf{1}_n$ is a $(\bar{q} - \underline{q} + 1)$ vector of zeros except for the n^{th} entry which equals 1.

5. Expected Profit Frontiers: Fine Tuning High Frequency Strategies

Here we show the tradeoff between expected profits and exposure to: volatility of terminal PnL ($\mathcal{R}^1$), volatility of terminal inventory ($\mathcal{R}^2$), and volatility of lifetime inventory ($\mathcal{R}^3$). Unless otherwise stated, in this section, the model parameters are: $T = 300$ seconds, $\kappa^\pm = 25$, $\lambda^\pm = 2$, $\sigma = 0.1$, $\bar{q} = -\underline{q} = 10$, $S_0 = 100$, and upon the arrival of a market order prices jump by $\epsilon^\pm = \frac{1}{25}$ with a probability of $\frac{1}{10}$ and do not jump with probability $\frac{9}{10}$.

5.1. Expected profit frontiers and risk metrics for High Frequency traders

HF traders seek to maximize expected profits but also take into account the tradeoff between profits and their exposure to inventory risk. In Figure 7 we depict three investment frontiers that show the risk-return profile for the three risk metrics developed above as the penalty parameter ϕ and the liquidation cost parameter α vary. To depict the frontiers we use the optimal strategy in Proposition 1 and calculate the risk metrics by numerical integrating the system of ODEs (A.3) and (A.6).⁸

Panels (a), (b), and (c) show the tradeoff between the mean of the PnL and the: volatility of the PnL, volatility of terminal inventory, and volatility of lifetime inventory respectively. The three panels show that increasing the inventory penalty parameter ϕ reduces both the mean PnL and the exposure to risk as measured by $\mathcal{R}_1$, $\mathcal{R}_2$, and $\mathcal{R}_3$. Moreover, it is interesting to see that there are ranges where fine tuning ϕ can significantly reduce the risk of the strategy without a significant reduction in expected profits.

The panels also show how the expected profit frontiers depend on the liquidation penalty α . In panels (a) and (c) we observe that the liquidation penalty has virtually no effect on the tradeoff between the mean of the PnL and both the volatility of the PnL and the volatility of lifetime inventory. On the other hand,

⁸Note that the various matrices and vectors appearing in these ODEs will be time-dependent when the optimal strategy from Proposition 1 is inserted – which is why numerical integration is necessary.

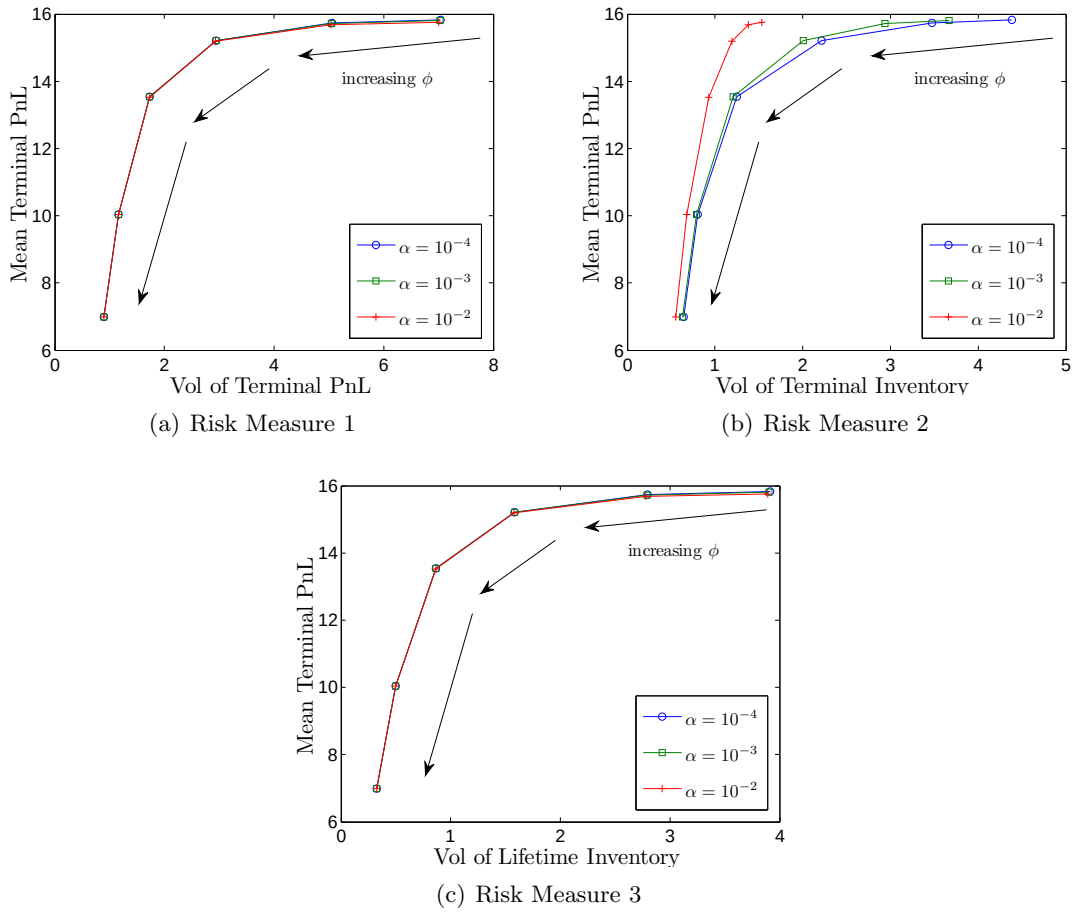


Figure 7: The effects of the penalty parameter ϕ and the terminal liquidation cost α on the risk-return profiles. Each point corresponds to $\phi = \{0, 0.01, 0.1, 1, 5, 10\}$. The investment horizon $T = 300$ seconds and the remaining model parameters are: $\kappa^{\pm} = 25$, $\lambda^{\pm} = 2$, $\epsilon^{\pm} = \frac{1}{25}\mathbf{a}^{\pm}$ where $\mathbf{a}^{\pm}$ are independent Bernoulli r.v. with success prob of $\frac{1}{10}$, $\sigma = 0.1$, $\bar{q} = -\underline{q} = 10$ and $S_0 = 100$.

panel (b) shows that the frontier that depicts the tradeoff between expected profits and the variance of final inventory is mildly affected by low values of the liquidation penalty, but when the penalty is increased to $\alpha = 0.01$ the frontier shifts to the left by an economically significant amount.⁹ A shift to the left implies that by increasing α (and not varying ϕ) the HF trader can achieve the same level of expected profits and considerably reduce her exposure to final inventory risk.

Overall, Figure 7 shows that HF traders who maximize terminal expected wealth and are constrained by inventories and capital requirements can fine tune their strategies to find an optimal point in the expected profit frontiers. The constraints are either self-imposed or required by a regulator or the exchange and the severity of these constraints will determine what is the HF trader's optimal choice.

⁹We note that for panels (a) and (c) the expected profit frontiers that result from employing the optimal limiting strategy in Proposition 6, and calculating the risk metric using the results in Proposition 8, coincide with the numerical solution of the optimal strategy in Proposition 1 presented here – the effect of the terminal date T on the expected profit frontiers is negligible.

5.2. Distribution of PnL, Terminal Inventory, and Lifetime Inventory

5.2.1. Symmetric arrival of market orders

Moreover, to further understand how HF traders can fine tune their strategies to maximize expected profits, whilst obeying risk controls, we employ 10,000 simulations¹⁰ and illustrate the effects that the penalties $\phi = \{0, 0.1, 2\}$ and $\alpha = \{0, 0.001, 0.01\}$ have on the distribution of the PnL, terminal inventory q_T , and inventory lifetime when the optimal strategy derived in Proposition 1 is employed. We recall that the model parameters are: $\lambda^\pm = 2$ (i.e. symmetric arrival of market orders), $\kappa^\pm = 25$, $\bar{q} = -\underline{q} = 10$, $\sigma = 0.1$, $T = 300$ seconds and upon the arrival of a market order prices jump by $\epsilon^\pm = \frac{1}{25}$ with a probability of $\frac{1}{10}$ and do not jump with probability $\frac{9}{10}$. The results are reported in Panel A of Table 1.

The table shows the mean and standard deviations of the PnL, lifetime inventory, and terminal inventory from following the optimal policy. We observe that for any level of liquidation cost α , the effect of an increase in the penalty ϕ reduces both the expected profit and its standard deviation.

It is also clear, that everything else equal, the HF trader will prefer strategies where the lifetime inventory is concentrated around the optimal level of inventories which in this case is zero because $\lambda^+ = \lambda^-$. In the table we see that penalizing asset holdings along the entire path of the strategy with $\phi = 2$ ensures that inventories revert to zero very quickly.

Panel A clearly shows that the higher is the liquidation penalty and the higher is the penalization of inventories throughout the entire strategy, the terminal inventory is more concentrated around zero. This information will be used by the HF trader to choose among strategies with similar PnL distributions by picking the ones that are least exposed to terminal inventory.

Thus, given the liquidation costs (which proxy for the price impact of large market orders as well as capturing the fees charged by the exchange for taking liquidity) the HF trader will fine tune her strategy by looking at Table 1 and picking a preferred ϕ and/or by looking at the expected profit. For example, an HF trader that is highly constrained by the amount of inventory that she can hold at any point in time, will select an HF strategy with a high ϕ at the expense of low expected profits – the HF trader must ensure that the expected profits are sufficiently high to allow her to recover the fixed and variable costs from being in the business.

Finally, we remark that the liquidation penalty α seems to be useful only to reduce the volatility of the final inventory. On the other hand, ϕ plays a prominent role in the distribution of the PnL, terminal inventory, and lifetime inventory. The inventory penalty parameter ϕ is self-imposed by the HF trader

¹⁰One path of the simulation is obtained by generating the time of the next market order (occurring at a rate of $\lambda^+ + \lambda^-$); generating a uniform random variable to decide if it is a buy/sell (it is a buy with probability $\frac{\lambda^+}{\lambda^+ + \lambda^-}$); generating a uniform random variable to decide if the limit order on the appropriate side of the book is filled (occurs with probability $e^{-\kappa^\pm \delta_t^{\pm}}$). Moreover, the cash and inventories are tracked and the process repeated until maturity at which point any remaining inventory is liquidated (accounting for the liquidation cost). It is important to note that this process effectively prevents the optimal spreads from being negative, because a negative spread implies that market orders arrive faster. However, here market orders always arrive at the specified rate. See 6 for further discussion of this and related points.

and captures how binding are the constraints on intraday inventory exposures for the HF trader which could be different depending on the market –and particular asset– she is trading.

Panel A									
Symmetric Market: $\lambda^+ = 2, \lambda^- = 2$									
	PnL			Lifetime Inventory			Terminal Inventory		
	$\phi = 0$	$\phi = 0.1$	$\phi = 2$	$\phi = 0$	$\phi = 0.1$	$\phi = 2$	$\phi = 0$	$\phi = 0.1$	$\phi = 2$
$\alpha = 0$	15.776 (7.10)	15.210 (2.97)	12.392 (1.52)	0.008 (3.92)	-0.000 (1.59)	-0.000 (0.74)	0.002 (4.75)	-0.017 (2.20)	-0.006 (0.95)
$\alpha = 0.001$	15.802 (7.13)	15.170 (2.94)	12.375 (1.54)	0.007 (3.89)	-0.000 (1.59)	-0.000 (0.74)	0.005 (3.54)	-0.017 (1.98)	-0.006 (0.93)
$\alpha = 0.01$	15.716 (7.02)	15.201 (2.96)	12.368 (1.52)	0.007 (3.85)	-0.000 (1.58)	-0.000 (0.74)	-0.008 (1.51)	0.001 (1.26)	-0.003 (0.80)
Panel B									
Asymmetric Market: $\lambda^+ = 1, \lambda^- = 3$									
	PnL			Lifetime Inventory			Terminal Inventory		
	$\phi = 0$	$\phi = 0.1$	$\phi = 2$	$\phi = 0$	$\phi = 0.1$	$\phi = 2$	$\phi = 0$	$\phi = 0.1$	$\phi = 2$
$\alpha = 0$	31.596 (15.58)	22.338 (7.52)	10.524 (1.54)	-8.640 (1.45)	-3.886 (1.61)	-0.127 (0.74)	-7.143 (1.89)	-1.523 (2.16)	0.259 (0.94)
$\alpha = 0.001$	31.547 (15.69)	22.415 (7.54)	10.566 (1.54)	-8.631 (1.46)	-3.883 (1.61)	-0.127 (0.74)	-6.023 (2.05)	-1.288 (1.96)	0.244 (0.93)
$\alpha = 0.01$	31.317 (15.70)	22.361 (7.49)	10.534 (1.52)	-8.592 (1.52)	-3.874 (1.61)	-0.128 (0.74)	-2.272 (1.63)	-0.620 (1.28)	0.163 (0.81)

Table 1: Mean and standard deviations of the PnL, lifetime inventory, and terminal inventory from following the optimal policy with $\kappa^\pm = 25$, $\epsilon^\pm = \frac{1}{25}\mathbf{a}^\pm$ where $\mathbf{a}^\pm$ are independent Bernoulli r.v. with success prob of $\frac{1}{10}$, $\bar{q} = -\underline{q} = 10$, $\sigma = 0.1$ and $T = 300$ seconds as the penalty parameter ϕ and liquidation cost parameter α vary.

5.2.2. Asymmetric arrival of market orders

Above we assumed that the arrival rate of market buy and sell orders was the same, on average there were 4 market orders per second and half of them were buy and the other half were sell. Here we proceed as above but assume that the arrival of market orders is asymmetric, $\lambda^+ = 1$ and $\lambda^- = 3$, so that on average there are four market orders per second but there are much more sell than buy orders (market momentum). We perform 10,000 simulations and the remaining parameters are as above: $\phi = \{0, 0.1, 2\}$, $\alpha = \{0, 0.001, 0.01\}$, $\kappa^\pm = 25$, $\bar{q} = -\underline{q} = 10$, $\sigma = 0.1$, $T = 300$ seconds and upon the arrival of a market order prices jump by $\epsilon^\pm = \frac{1}{25}$ with a probability of $\frac{1}{10}$ and do not jump with probability $\frac{9}{10}$. We report the results in Panel B of Table 1.

Panel B shows that when inventories are not penalized throughout the entire strategy, the mean PnL in the asymmetric market is higher than the mean PnL resulting from a symmetric market. This occurs because in the former the strategy benefits from roundtrip trades as well as inventory appreciation. When the strategy is far away from expiry it is optimal to quickly build a short inventory position, and then

start executing roundtrips around that short optimal level. The excess arrival of market sell orders exerts a downward pressure on the midquote which means that on average the HF trader earns a profit on short positions.

Panel B also shows that even though the HF trader can profit from the expected appreciation of her short position, if $\phi = 0.1$ her inventory levels are ‘forced’ to be closer to zero and these appreciation gains are forgone. Furthermore, if the penalization of inventory is further increased to $\phi = 2$, then the distribution of the PnL in the asymmetric market shifts to the left of that of a symmetric market. The intuition is that in an asymmetric market the expected appreciation gains are an important component of the PnL, but these are considerably curbed by ϕ .

Figure 8 shows histograms of lifetime inventory, column (a), and terminal inventories, column (b), for both the symmetric (grey bars) and asymmetric (white bars) markets. Column (a) shows that when the penalty parameter ϕ is increased, the distribution of lifetime inventory becomes more symmetric and centered at zero. When $\phi = 0$ the optimal strategy profits from the expected decline in prices by building a short position in the asset and then performing as many roundtrips as possible, but always reverting to an optimal short inventory position – the strategy reaches the minimum allowed $q = -10$ and spends most of the time between $q = -9$ and $q = -10$. As the inventory penalties are increased, the optimal strategy relies less on gains via inventory appreciation and more on roundtrips with inventories reverting to zero.

Column (b) shows that when $\phi = 0$ the distribution of terminal inventory is negatively skewed and to the left of that of the symmetric market. As the penalty parameter ϕ is increased the distribution of terminal inventory becomes skewed to the right. Initially this result may seem counterintuitive because one would expect that in a market with negative momentum it is always optimal to keep a short position which is either unwound fractions of a second before T or liquidated at time T . But on the contrary, the figure shows that the histogram is positively skewed and that the mean final inventory level is also positive. To understand the intuition behind this result assume that we are at expiry or milliseconds before expiry. In this case the chances of completing a roundtrip trade are essentially zero. Thus, the HF trader knows that if anything she can only execute the first leg of a trade to then liquidate it at the midprice and pick up the liquidation penalty. If we assume, for simplicity, that the liquidation penalty is $\alpha = 0$ and we have reached expiry, the best strategy is to maximize the probability of being filled on either side of the book – the same argument that follows still applies if the strategy is microseconds before expiry. In this case the optimal deltas are given by $\delta^\pm = \delta = 1/\kappa + \varepsilon$ (see Proposition 1) and the fill rates are $\Lambda^\pm = \lambda^\pm e^{-\kappa\delta}$. Obviously $\Lambda^- > \Lambda^+$, because $\lambda^- > \lambda^+$, and one can verify that although at the beginning of the strategy it is optimal to build short positions to gain from the expected fall in prices (see panel (b) in Figure 5), as we approach expiry the optimal postings will be such that the HF trader is more likely to be buying rather than selling the asset. This explains why we see the positive skew in the distribution shown in the last picture of column (b) Figure 8 for $\phi = 2$. We remark that although this effect is always present at times close to expiry, in our examples it only becomes visible in the last picture of column (b) occurs because a high ϕ ‘forces’ the strategy not to deviate from zero inventories before T , but at the last moment the optimal limit orders are more likely to be filled by sell, rather than buy, market orders. Therefore, at the end of the strategy there is an upward pressure on inventories and the distribution of terminal inventory is shifted and skewed to the right.

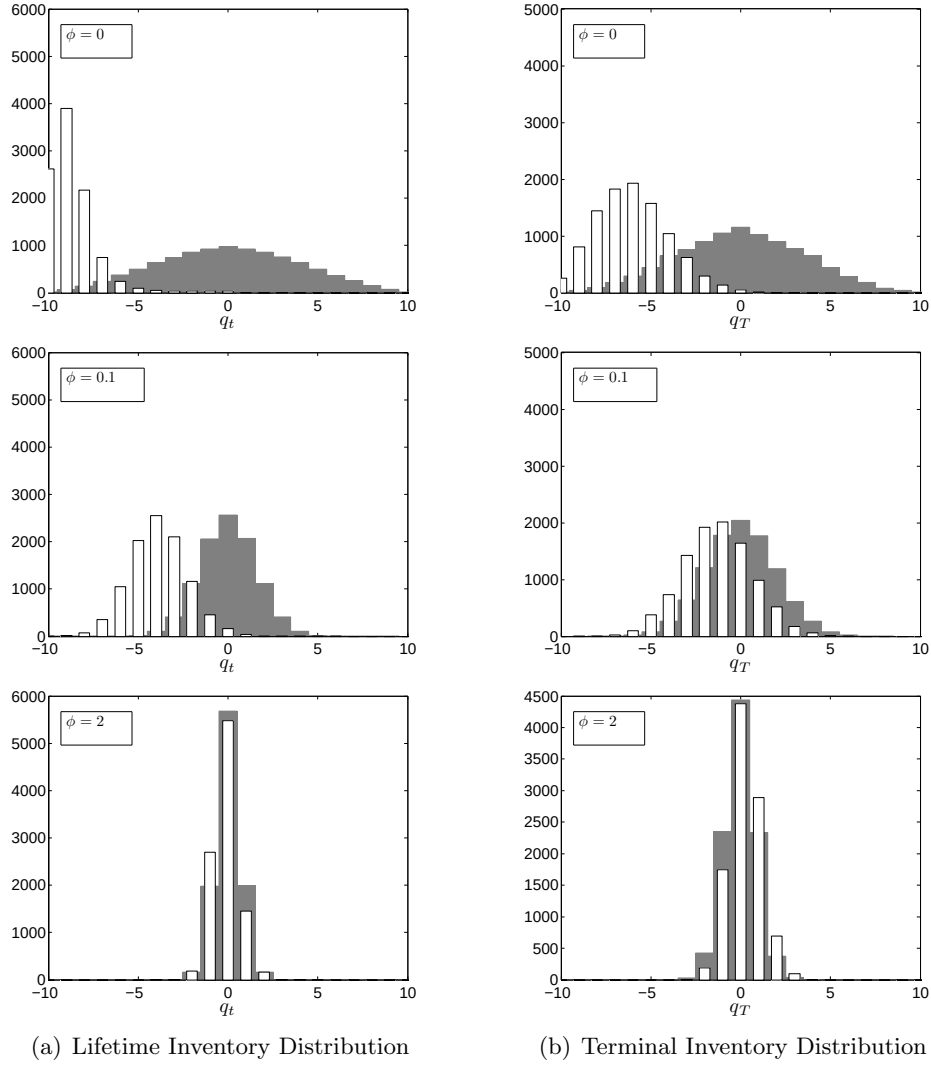


Figure 8: Distribution of inventory using lifetime inventory and terminal inventory levels from following the optimal policy with $\lambda^+ = 1$, $\lambda^- = 3$ (white bars) and $\lambda^\pm = 2$ (grey bars), $\kappa^\pm = 25$, $\epsilon^\pm = \frac{1}{25} \mathbf{a}^\pm$ where $\mathbf{a}^\pm$ are independent Bernoulli r.v. with success prob of $\frac{1}{10}$, $\bar{q} = -\underline{q} = 10$, $\sigma = 0.1$, $T = 300$ seconds, $\alpha = 0.001$ and $\phi = 0, 0.1$ and 2 (from top to bottom).

Finally, if we look at the last pictures of columns (a) and (b) we corroborate the intuition given above to explain the positive skew in terminal inventories. To this end, note that although the distribution of lifetime inventory is left skewed, the distribution of the final inventory is skewed to the right, which confirms that very close to and at expiry the optimal strategy is to increase inventories as much as possible – this last upward pressure on inventories is the same for all cases, but only becomes visible for high ϕ .

6. Constrained Strategies

As a final note, we discuss the issue of negative spreads ($\delta_t^\pm < 0$). Recall that conditional on the arrival of a market order, the probability of a limit order at price $S_t \pm \delta_t^\pm$ being filled is assumed to be $e^{-\kappa^\pm \delta_t^\pm}$ resulting in a rate of arrival of filled limit orders of $\Lambda^\pm = \lambda^\pm e^{-\kappa^\pm \delta_t^\pm}$. Consequently, negative spreads

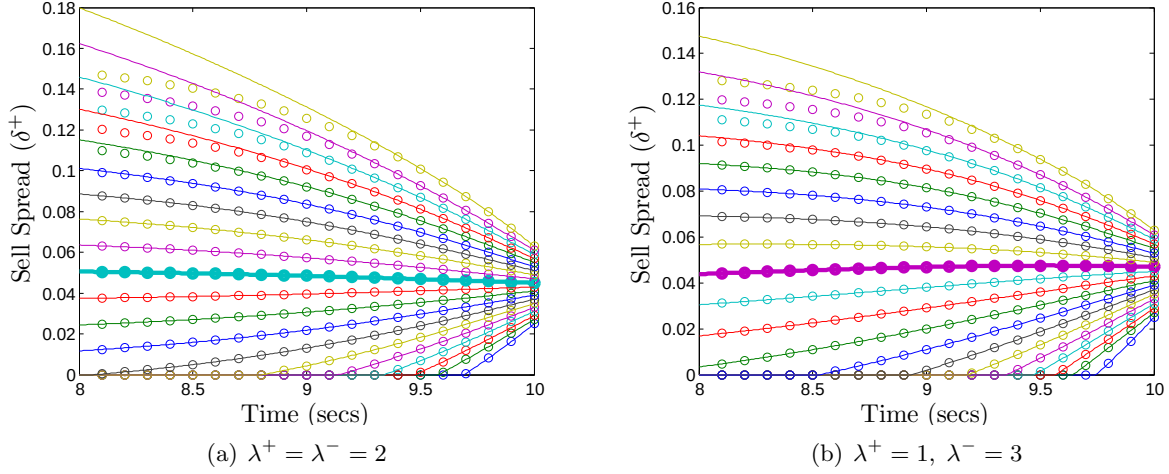


Figure 9: Comparison between the exact constrained spreads (solid lines) and the spreads computed by constraining afterwards (circles). The remaining model parameters are $T = 10$, $\alpha = 0.001$, $\phi = 0.1$, $\kappa^+ = \kappa^- = 25$, $\varepsilon^+ = \varepsilon^- = \frac{1}{250}$, $\bar{q} = -\underline{q} = 10$ and $\sigma = 0.1$. The inventories decrease from 10 to -9 from the smallest spreads to the largest. The thick lines show the spreads at the mean-reverting inventory level: top panels $q = 0$, bottom panels $q = -1$.

$(\delta_t^\pm < 0)$ can be interpreted as increasing the rate of arrival of market orders. However, this may not be a desirable feature. To correct this effect we can restrict the spreads to be non-negative. In this case, the optimal controls are given by

$$\delta^+(t, q) = \max \left(\frac{1}{\kappa^+} + \varepsilon^+ + \alpha(1 - 2q) - h_{q-1}(t) + h_q(t), 0 \right), \quad q \neq -\underline{q}, \quad \text{and} \quad (24a)$$

$$\delta^-(t, q) = \max \left(\frac{1}{\kappa^-} + \varepsilon^- + \alpha(1 + 2q) - h_{q+1}(t) + h_q(t), 0 \right) \quad q \neq \bar{q}, \quad (24b)$$

rather than (8). Upon substituting the feedback controls into the HJB equation the resulting coupled system of ODEs no longer admits a closed-form solution. However, it is straightforward to numerically integrate this system.

Figure 9 compares the spreads computed by numerically solving the ODEs to the spreads computed by ignoring the constraint but then replacing $\delta_t^\pm \rightarrow \max(\delta_t^\pm, 0)$ after the fact. This after-the-fact truncation of spreads to non-negative values is how our simulations are carried out. Notice that for most of the inventory levels corresponding to small/moderate spreads the exact constrained spreads and the spreads constrained afterwards are essentially identical. Only at large spreads, corresponding to short inventory levels on the sell side of book, are the solutions different. However, these inventories are only rarely attained and the impact on the overall strategy appears to be minimal. The highlighted spreads correspond to the mean-reverting inventory levels – measured by the highest lifetimes of inventory. Only the sell side is shown here, the buy side is similar except that $\bar{q}$ and $-\underline{q}$ are interchanged in the sell side analog to Figure 9(a) – as required by Corollary 3 – while in the analog to Figure 9(b), the symmetry is only approximate.

7. Conclusions

We propose risk metrics for HF trading strategies that profit from market making at very high speeds. The HF trader maximizes expected terminal wealth, and is capital and inventory constrained. These constraints enter into the optimization problem by penalizing accumulated inventories throughout the entire strategy, imposing a financial penalty for liquidating terminal inventories, and by imposing a cap on the amount of accumulated inventories that the HF trader can hold, long or short, at any point in time.

The risk metrics calculate the variance of terminal PnL, variance of terminal inventory, and the variance of lifetime inventory. We show that these metrics provide information that enables HF traders to fine tune their strategies to trade off expected profits against exposure to different metrics of inventory risk (which also proxy for capital constraints). We remark that the risk metrics we propose are for HF trading strategies in general and not limited to the particular ones discussed here.

For the optimal strategies we develop here, our analysis shows that final inventory penalties (which proxies the cost of unwinding positions at expiry of the strategy) affect the distribution of inventories at the expiry of the strategy, but has a very limited effect on the distribution of the PnL of the strategy. On the other hand, penalizing inventories throughout the entire strategy (the penalization affects inventory management but does not alter the PnL directly) has an important effect on the three risk metrics we propose. For example, we show that there are ranges of the penalization parameter where a very small increase in the parameter causes a considerable reduction in the risks borne by the HF trader (as given by our risk metrics), whilst expected profits are hardly affected.

Furthermore, the strategies include a component to recover adverse selection costs and induce mean reversion of inventories to an optimal level. This optimal level can change sign throughout the life of the strategy. When markets are asymmetric (arrival of sell market orders is different from that of buy orders) the optimal strategy benefits from both inventory appreciation and roundtrip trades. In a market with positive momentum – the arrival rate of buy market orders is higher than the arrival rate of market sell orders – the optimal strategy starts by quickly building positive inventories and is immediately followed by executing roundtrip trades whilst reverting to a time-dependent positive inventory level. However, as expiry is approached, the optimal postings are such that the optimal inventory level decreases and changes sign from positive to negative. This happens because very close to or at expiry the chances of executing roundtrip trades are virtually zero, and the expected gains from inventory appreciation (due to market momentum) are also very low, so the optimal strategy is to maximize the probability of being filled, and then liquidate the position, which in a market with positive momentum means that the optimal inventory level very close to or at expiry is negative.

Finally, we also discuss other possible risk metrics that are closely connected to the risk metrics we propose. In particular we show that the distribution of the lifetime inventory provides important information for HF traders who are capital and inventory constrained. For example, everything else equal, an HF trader prefers a strategy where inventories are turned over very quickly but the lifetime inventory is highest for the optimal level of inventory (which is not necessarily zero as discussed above). Thus, to provide further

insights into the fine tuning of the optimal strategies derived here, we show the trade off between expected profits and the distribution of lifetime inventory as a function of the parameter ϕ that penalizes inventories throughout the entire strategy. In particular, we show that small increases in ϕ have an important effect on the distribution of lifetime inventory whilst the distribution of the PnL (expected profits) is much less affected by this parameter – this provides a useful tool to fine tune HF strategies.

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Appendix A. Proofs

Appendix A.1. Proof of Proposition 2

Inserting the controls (8) into the HJB equation (7) and writing $h_q(t) = \frac{1}{\kappa} \ln \omega_q(t)$, after some straightforward computations, one finds that $\omega_q(t)$ satisfy the coupled system of equations

$$\partial_t \omega(t) + \mathbf{A} \omega(t) = \mathbf{0}. \quad (\text{A.1})$$

The boundary condition $h_q(T) = 0$ translates into $\omega_q(T) = 1$. This is a first order coupled system of ODEs. The solution to (A.1) is trivially given by (10).

Appendix A.2. Proof of Corollary 3

Let $\tilde{\omega}_q(t) = e^{\kappa \alpha q^2} \omega_q(t)$, then through straightforward computations we see that $\partial_t \tilde{\omega} + \tilde{\mathbf{A}} \tilde{\omega} = \mathbf{0}$ subject to $\tilde{\omega}_q(T) = e^{\kappa \alpha q^2}$ where

$$\tilde{\mathbf{A}}_{i,q} = \begin{cases} -\phi \sigma^2 \kappa q^2, & i = q, \\ \lambda e^{-\kappa(\frac{1}{\kappa} + \varepsilon)}, & i = q-1, q+1 \\ 0, & \text{otherwise.} \end{cases}$$

Clearly, this system is symmetric under $q \rightarrow -q$, therefore $\tilde{\omega}_q(t) = \tilde{\omega}_{-q}(t)$. Furthermore, $\omega_q(t) = \omega_{-q}(t)$ and hence $h_q(t) = h_{-q}(t)$. Finally, the optimal feedback control $\delta^{+*}(t, q) = \frac{1}{\kappa} + \varepsilon + \alpha(1 - 2q) - h_{q-1}(t) + h_q(t) = \frac{1}{\kappa} + \varepsilon + \alpha(1 + 2(-q)) - h_{-(q-1)}(t) + h_{-q}(t) = \delta^{-*}(t, -q)$. $\square$

Appendix A.3. Proof of Proposition 6

Part 1: Clearly, $e^{\mathbf{A}(T-t)} = \mathbf{U} e^{\mathbf{D}(T-t)} \mathbf{U}^{-1}$ where $\mathbf{D}$ is the diagonal matrix of eigenvalues of $\mathbf{V}$. We then have $\exp\{\mathbf{A}(T-t)\} \mathbf{1} = \mathbf{U} \exp\{[\mathbf{D}]_\ell(T-t)\} \mathbf{U}^{-1} \mathbf{1} + o(\exp\{\mathbf{D}_{\ell\ell}(T-t)\})$, where $[\mathbf{D}]_\ell$ denotes the diagonal matrix whose entries are all zero except for the ℓ^{th} entry. Next, notice that $\exp\{[\mathbf{D}]_\ell(T-t)\} \mathbf{U}^{-1} \mathbf{1} = [\mathbf{U}^{-1} \mathbf{1}]_\ell e^{\mathbf{D}_{\ell\ell}(T-t)}$. Consequently, $\mathbf{h}(t) = \frac{\mathbf{D}_{\ell\ell}}{\kappa}(T-t) + \frac{1}{\kappa} \ln(\mathbf{U}[\mathbf{U}^{-1} \mathbf{1}]_\ell) + o(T-t)$ and the result immediately follows since the linear term is independent of q and applying Proposition 1.

Part 2: We have $e^{\mathbf{A}(T-t)} = \mathbb{I} + \mathbf{A}(T-t) + o(T-t)$, therefore $\omega(t) = \mathbf{1} + \mathbf{A} \mathbf{1}(T-t) + o(T-t)$ and $\mathbf{h}(t) = \frac{1}{\kappa} \mathbf{A} \mathbf{1}(T-t) + o(T-t)$. Moreover, for $q \neq \bar{q}, \underline{q}$ we have

$$(\mathbf{A} \mathbf{1})_q = \lambda^+ e^{-\kappa(\frac{1}{\kappa} + \varepsilon^+ - \alpha(1+2q))} + \lambda^- e^{-\kappa(\frac{1}{\kappa} + \varepsilon^- + \alpha(2q-1))} + q\kappa(\lambda^+ \epsilon^+ - \lambda^- \epsilon^-) - \phi \sigma^2 \kappa q^2.$$

Using Proposition 1, after some tedious computations we arrive at the result.

Appendix A.4. Proof of Proposition 8

Part 1: Since the rows of $\mathbf{B}$ sum to zero, and non-diagonal entries are positive, it represents a generator matrix of a Markov chain. Consequently, $\mathbf{1}$ is an eigenvector with eigenvalue zero and its remaining eigenvalues are negative. This proves part 1.

Part 2: We compute $M_1(t, x, q, S_t) \triangleq \mathbb{E}_t[X_T + q_T(S_T - \alpha q_T)]$ and $M_2(t, x, q, S_t) \triangleq \mathbb{E}_t[(X_T + q_T(S_T - \alpha q_T))^2]$ under the strategy δ_t^* separately. Clearly, $M_j(t, X_t, q_t, S_t)$ are martingales under any strategy, therefore they must each satisfy the following PIDE:

$$\begin{aligned} 0 = & \partial_t M_j + \frac{1}{2} \sigma^2 \partial_{SS} M_j \\ & + \lambda^+ \left\{ e^{-\kappa \delta^{*+}(q)} \mathbb{E} \left(M_j(t, x + S + \delta^{*+}(q), q - 1, S + \epsilon^+) - M_j(t, x, q, S) \right) \right. \\ & \quad \left. + \left(1 - e^{-\kappa \delta^{*+}(q)} \right) \mathbb{E} \left(M_j(t, x, q, S + \epsilon^+) - M_j(t, x, q, S) \right) \right\} \\ & + \lambda^- \left\{ e^{-\kappa \delta^{*-}(q)} \mathbb{E} \left(M_j(t, x - S + \delta^{*-}(q), q + 1, S - \epsilon^-) - M_j(t, x, q, S) \right) \right. \\ & \quad \left. + \left(1 - e^{-\kappa \delta^{*-}(q)} \right) \mathbb{E} \left(M_j(t, x, q, S - \epsilon^-) - M_j(t, x, q, S) \right) \right\}, \end{aligned} \quad (\text{A.2a})$$

for $q = \underline{q}, \underline{q} + 1, \dots, \bar{q}$. Here, and in the remainder of this proof, we have written $\delta^{*+}(\underline{q}) = \delta^{*-}(\bar{q}) = +\infty$ to avoid having to treat these boundary cases separately.

Solving for M_1 : Using the ansatz $M_1 = x + q(S - \alpha q) + m_1(t, q)$ then after some tedious computations, $m_1(q)$ satisfies

$$\partial_t \mathbf{m}_1 + \mathbf{B} \mathbf{m}_1 + \boldsymbol{\theta} = \mathbf{0}, \quad (\text{A.3})$$

where the vector $\boldsymbol{\theta}$ has elements

$$\begin{aligned} \theta_q = & \left(\varepsilon^+ \tilde{\lambda}^+ - \varepsilon^- \tilde{\lambda}^- \right) q + e^{-\kappa \delta^{*+}(q)} \tilde{\lambda}^+ \left(\delta^{*+}(q) - \varepsilon^+ - \alpha(1 - 2q) \right) \\ & + e^{-\kappa \delta^{*-}(q)} \tilde{\lambda}^- \left(\delta^{*-}(q) - \varepsilon^- - \alpha(1 + 2q) \right) \end{aligned} \quad (\text{A.4})$$

subject to $\mathbf{m}_1(T) = \mathbf{0}$. Delicate treatment of the zero eigenvalue of the matrix $\mathbf{B}$ is required, but it is not difficult to show that the solution to (A.3) is given by $\mathbf{m}_1 = \mathbf{V} \tilde{\mathbf{m}}_1$ where the components of $\tilde{\mathbf{m}}_1$ are

$$\tilde{\mathbf{m}}_{1,j} = \begin{cases} (\mathbf{V}^{-1} \boldsymbol{\theta})_1 \tau, & j = 1, \\ (\mathbf{V}^{-1} \boldsymbol{\theta})_j \frac{e^{d_j \tau} - 1}{d_j}, & j > 1. \end{cases} \quad (\text{A.5})$$

Solving for M_2 : Using the ansatz $M_2 = (x + q(S - \alpha q))^2 + m_2(t, x, q, S)$ with $m_2(t, x, q, S) = a(t, q) + b(t, q)x + c(t, q)S$, we find that $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ satisfy the coupled system of equations

$$\partial_t \mathbf{a} + \mathbf{B} \mathbf{a} + \boldsymbol{\Upsilon} \mathbf{b} + \boldsymbol{\Omega} \mathbf{c} + \mathbf{v} = \mathbf{0}, \quad (\text{A.6a})$$

$$\partial_t \mathbf{b} + \mathbf{B} \mathbf{b} + \boldsymbol{\chi} = \mathbf{0}, \quad (\text{A.6b})$$

$$\partial_t \mathbf{c} + \mathbf{B} \mathbf{c} + \boldsymbol{\Psi} \mathbf{b} + \boldsymbol{\psi} = \mathbf{0}, \quad (\text{A.6c})$$

where the elements of the matrices Υ , Ω and Ψ are

$$\Upsilon_{i,q} = \begin{cases} e^{-\kappa \delta^{*+}(q)} \tilde{\lambda}^+ \delta^{*+}(q), & i = q - 1, \\ e^{-\kappa \delta^{*-}(q)} \tilde{\lambda}^- \delta^{*-}(q), & i = q + 1, \\ 0 & \text{otherwise,} \end{cases} \quad (\text{A.7a})$$

$$\Omega_{i,q} = \begin{cases} \left\{ \left(1 - e^{-\kappa \delta^{*+}(q)} \right) \tilde{\lambda}^+ - \left(1 - e^{-\kappa \delta^{*-}(q)} \right) \tilde{\lambda}^- \right\} \varepsilon, & i = q, \\ e^{-\kappa \delta^{*+}(q)} \tilde{\lambda}^+ \varepsilon, & i = q - 1, \\ -e^{-\kappa \delta^{*-}(q)} \tilde{\lambda}^- \varepsilon, & i = q + 1, \\ 0 & \text{otherwise,} \end{cases} \quad (\text{A.7b})$$

$$\Psi_{i,q} = \begin{cases} e^{-\kappa \delta^{*+}(q)} \tilde{\lambda}^+, & i = q - 1, \\ -e^{-\kappa \delta^{*-}(q)} \tilde{\lambda}^-, & i = q + 1, \\ 0 & \text{otherwise,} \end{cases} \quad (\text{A.7c})$$

and the vectors $\mathbf{v}$, $\mathbf{\chi}$ and $\mathbf{\Psi}$ have elements

$$\mathbf{v}_q = \tilde{\lambda}^+ \left\{ q^2 (\varepsilon_2^+ - 2\alpha q \varepsilon^+) + e^{-\kappa \delta^{*+}} \left(\hat{\delta}^+(q) + 2\alpha q \right) \left(\hat{\delta}^+(q) - 2\alpha (q^2 - q) + 2q \varepsilon^+ \right) \right\} \\ + \tilde{\lambda}^- \left\{ q^2 (\varepsilon_2^- + 2\alpha q \varepsilon^-) + e^{-\kappa \delta^{*-}} \left(\hat{\delta}^-(q) - 2\alpha q \right) \left(\hat{\delta}^-(q) - 2\alpha (q^2 + q) - 2q \varepsilon^- \right) \right\}, \quad (\text{A.8a})$$

$$\mathbf{\chi}_q = 2\tilde{\lambda}^+ \left\{ (\hat{\delta}^+(q) + 2\alpha q) e^{-\kappa \delta^{*+}(q)} + \varepsilon^+ q \right\} + 2\tilde{\lambda}^- \left\{ (\hat{\delta}^-(q) - 2\alpha q) e^{-\kappa \delta^{*-}(q)} - \varepsilon^- q \right\}, \quad (\text{A.8b})$$

$$\mathbf{\psi}_q = 2\tilde{\lambda}^+ \left\{ q \left(\hat{\delta}^+(q) + 2\alpha q \right) e^{-\kappa \delta^{*+}(q)} + q^2 \varepsilon^+ \right\} + 2\tilde{\lambda}^- \left\{ q \left(\hat{\delta}^-(q) - 2\alpha q \right) e^{-\kappa \delta^{*-}(q)} - q^2 \varepsilon^- \right\}, \quad (\text{A.8c})$$

where

$$\hat{\delta}^\pm(q) = \delta^{*\pm}(q) - \alpha - \varepsilon^\pm, \quad \text{and} \quad \varepsilon_2^\pm = \mathbb{E}[(\varepsilon^\pm)^2]. \quad (\text{A.8d})$$

The existence of the zero eigenvalue in the matrix $\mathbf{B}$ introduces some complexity in the solution to ODEs. This system of ODEs is solved by (i) solving (A.6b) by diagonalization (using the eigenvectors of $\mathbf{B}$). The solution to the equations corresponding to non-zero(negative) eigenvalues decays exponentially. The solution to the equations corresponding to the zero eigenvalue grows linearly. (ii) Substituting the solution for $\mathbf{b}$ into (A.6c), diagonalizing once again and carefully separating out the zero-eigenvalues which now interact with the remaining eigenvalues. The zero eigenvalue solution now contains linear combinations of τ , τ^2 and $e^{-d_k \tau}$, while the remaining eigenvalue solutions are linear combinations of τ , $\tau e^{-d_k \tau}$ and $e^{-d_k \tau}$. (iii) Substituting the solution for both $\mathbf{b}$ and $\mathbf{c}$ into (A.6a), diagonalizing again and treating the interaction of the zero and non-zero eigenvalues. The zero eigenvalue solution is now a linear combination of τ , τ^2 , τ^3 , $e^{-d_k \tau}$ and $\tau e^{-d_k \tau}$, while the non-zero eigenvalue solutions are linear combinations of τ , τ^2 , $e^{-d_k \tau}$, $\tau e^{-d_k \tau}$ and $\tau^2 e^{-d_k \tau}$. After some tedious computations we find that

$$\mathbf{a} = \mathbf{V} \tilde{\mathbf{a}}, \quad \mathbf{b} = \mathbf{V} \tilde{\mathbf{b}}, \quad \mathbf{c} = \mathbf{V} \tilde{\mathbf{c}}, \quad (\text{A.9})$$

where,

$$\tilde{\mathbf{a}}_j = \begin{cases} i_1 \tau + m_1 \tau^2 + r_1 \tau^3 + \sum_{k \neq 1} n_{1k} (e^{d_k \tau} - 1) + \sum_{k \neq 1} o_{1k} \tau e^{d_k \tau}, & j = 1, \\ i_j \tau + m_j \tau^2 + h_j (1 - e^{d_j \tau}) \\ + \sum_{k \neq j, 1} n_{jk} (e^{d_k \tau} - e^{d_j \tau}) + \sum_{k \neq j, 1} o_{jk} \tau e^{d_k \tau} + q_j \tau^2 e^{d_j \tau}, & j > 1, \end{cases} \quad (\text{A.10a})$$

$$\tilde{\mathbf{b}}_j = \begin{cases} \tilde{\chi}_1 \tau, & j = 1, \\ \tilde{\chi}_j \frac{e^{d_j \tau} - 1}{d_j}, & j > 1, \end{cases} \quad (\text{A.10b})$$

$$\tilde{\mathbf{c}}_j = \begin{cases} -\Theta_1 \tau + \frac{1}{2} \tilde{\Psi}_{11} \tilde{\chi}_1 \tau^2 + \sum_{k \neq 1} \frac{\hat{\Psi}_{1k}}{d_k} (e^{d_k \tau} - 1), & j = 1, \\ -\Theta_j \tau + \eta_j \frac{e^{d_j \tau} - 1}{d_j} + \sum_{k \neq j, 1} \frac{\hat{\Psi}_{jk}}{d_k - d_j} (e^{d_k \tau} - e^{d_j \tau}) + \hat{\Psi}_{jj} \tau e^{d_j \tau}, & j > 1, \end{cases} \quad (\text{A.10c})$$

and the various coefficients are given by

$$\begin{aligned} \tilde{\chi}_j &= (\mathbf{V}^{-1} \boldsymbol{\chi})_j, & \tilde{\psi}_j &= (\mathbf{V}^{-1} \boldsymbol{\psi})_j, & \tilde{v}_j &= (\mathbf{V}^{-1} \mathbf{v})_j, \\ \tilde{\Psi}_{jk} &= (\mathbf{V}^{-1} \boldsymbol{\Psi} \mathbf{V})_{jk}, & \tilde{\Omega}_{jk} &= (\mathbf{V}^{-1} \boldsymbol{\Psi} \boldsymbol{\Omega})_{jk}, & \tilde{\Upsilon}_{jk} &= (\mathbf{V}^{-1} \boldsymbol{\Psi} \boldsymbol{\Upsilon})_{jk}, \\ \hat{\Psi}_{jk} &= \frac{\tilde{\Psi}_{jk} \tilde{\chi}_k}{d_k}, & \hat{\Omega}_{jk} &= \frac{\tilde{\Omega}_{jk} \eta_k^c}{d_k}, & \hat{\Upsilon}_{jk} &= \frac{\tilde{\Upsilon}_{jk} \tilde{\chi}_k}{d_k}, \\ \bar{\Omega}_{jk} &= \tilde{\Omega}_{jk} \hat{\Psi}_{kk}, \end{aligned} \quad (\text{A.11a})$$

$$\Theta_j = \begin{cases} \sum_{k \neq 1} \hat{\Psi}_{1k} - \tilde{\psi}_1, & j = 1, \\ \frac{\tilde{\Psi}_{j1} \tilde{\chi}_1}{d_j}, & j > 1, \end{cases} \quad (\text{A.11b})$$

$$\eta_j = \tilde{\psi}_j + \Theta_j - \sum_{k \neq 1} \hat{\Psi}_{jk}, \quad (\text{A.11c})$$

$$i_j = (2m_j - \beta_j)/d_j, \quad m_j = \frac{1}{2} \tilde{\Omega}_{j1} \tilde{\Psi}_{11} \tilde{\chi}_1 / d_j, \quad h_j = (i_j - \delta_j)/d_j, \quad (\text{A.11d})$$

$$\delta_j = \tilde{v}_j - \sum_{k \neq 1} \zeta_{jk}, \quad \zeta_{jk} = \frac{1}{d_k} \left(\tilde{\Upsilon}_{jk} \tilde{\chi}_1 + \tilde{\Omega}_{jk} \eta_k + \tilde{\Omega}_{j1} \hat{\Psi}_{1k} \right), \quad (\text{A.11e})$$

$$\beta_j = \tilde{\Upsilon}_{j1} \tilde{\chi}_1 - \sum_{k \neq 1} \tilde{\Omega}_{jk} \Theta_k, \quad \xi_{jk} = \zeta_{jk} - \sum_{l \neq k, 1} (g_{kl} - g_{lk}), \quad g_{kl} = \sum_{m \neq l, 1} \frac{\tilde{\Omega}_{kl} \tilde{\Psi}_{lm}}{d_m - d_l}, \quad (\text{A.11f})$$

$$n_{jk} = \begin{cases} \frac{\xi_{1k} - o_{1k}}{d_k}, & j = 1, \\ (\xi_{jk} - o_{jk})/(d_k - d_j), & j \neq 1, \end{cases} \quad o_{jk} = \begin{cases} \tilde{\Omega}_{1k} \hat{\Psi}_{kk}/d_k, & j = 1, \\ \xi_{jj} + d_j n_{jj}, & k = j \neq 1, \\ \tilde{\Omega}_{jk} \hat{\Psi}_{kk}/(d_k - d_j), & k \neq j, 1, \end{cases} \quad (\text{A.11g})$$

$$q_j = \frac{1}{2} \tilde{\Omega}_{jj} \hat{\Psi}_{jj}, \quad r_1 = \frac{1}{6} \tilde{\Omega}_{11} \tilde{\Psi}_{11} \tilde{\chi}_1. \quad (\text{A.11h})$$

Large T asymptotics: Since the matrix $\mathbf{B}$ is a generator matrix of a Markov chain, its non-zero eigenvalues are negative. Consequently, for large τ , only the polynomial terms¹¹ of the functions $\tilde{\mathbf{a}}$, $\tilde{\mathbf{b}}$ and $\tilde{\mathbf{c}}$ contribute. This leads to the stated result.

¹¹Although only these terms contribute for large τ , it is necessary to construct the full solution because the boundary conditions depend on the exponential terms and feed into the constant term of the polynomial.

Part 3: For risk metric $\mathcal{R}^2$, note that $\mathbb{E}_t[q_T^2]$ and $\mathbb{E}_t[q_T]$ are given by Proposition 9 with $\ell = 0$ and $\varphi(q) = q^2$ and q , respectively. Consequently, we have

$$\mathcal{R}^2(t, \cdot) = \mathbf{V} \begin{pmatrix} (\mathbf{V}^{-1}\boldsymbol{\beta})_1 \\ e^{\mathbf{d}_2(T-t)}(\mathbf{V}^{-1}\boldsymbol{\beta})_2 \\ \vdots \\ e^{\mathbf{d}_{\bar{q}-\underline{q}+1}(T-t)}(\mathbf{V}^{-1}\boldsymbol{\beta})_{\bar{q}-\underline{q}+1} \end{pmatrix} - \left(\mathbf{V} \begin{pmatrix} (\mathbf{V}^{-1}\boldsymbol{\gamma})_1 \\ e^{\mathbf{d}_2(T-t)}(\mathbf{V}^{-1}\boldsymbol{\gamma})_2 \\ \vdots \\ e^{\mathbf{d}_{\bar{q}-\underline{q}+1}(T-t)}(\mathbf{V}^{-1}\boldsymbol{\gamma})_{\bar{q}-\underline{q}+1} \end{pmatrix} \right)^{\odot 2}. \quad (\text{A.12})$$

Here, the notation $A^{\odot 2}$ represents componentwise squaring. The limiting risk metric (as $T \rightarrow +\infty$) in (22) immediately follows from the fact that the non-zero eigenvalues are negative.

Part 4: Finally, for $\mathcal{R}^3(t, q)$ note that $\mathbb{E}_t \left[\int_t^T \mathbb{I}\{q_s = n\} ds \right]$ is given by Proposition 9 with $\ell(q) = \mathbb{I}\{q = n\}$ and $\varphi(q) = 0$. Consequently, we have,

$$\mathbb{Q}_n(t, q) = \frac{1}{T-t} \mathbf{V} \begin{pmatrix} (T-t)(\mathbf{V}^{-1}\mathbf{1}_n)_1 \\ \frac{e^{\mathbf{d}_2(T-t)}-1}{\mathbf{d}_2}(\mathbf{V}^{-1}\mathbf{1}_n)_2 \\ \vdots \\ \frac{e^{\mathbf{d}_{\bar{q}-\underline{q}+1}(T-t)}-1}{\mathbf{d}_{\bar{q}-\underline{q}+1}}(\mathbf{V}^{-1}\mathbf{1}_n)_{\bar{q}-\underline{q}+1} \end{pmatrix}. \quad (\text{A.13})$$

Once again, the limiting case (as $T \rightarrow +\infty$) in (23) follows from the fact that the non-zero eigenvalues are negative. $\square$

Proposition 9. Let $(\delta_t^{\pm*})_{0 \leq t \leq T}$ denote the limiting optimal spreads¹² given by Proposition 6 and $(q_t^*)_{0 \leq t \leq T}$ the corresponding inventory. Define, for arbitrary bounded functions $\varphi(q)$ and $\ell(q)$ for $q \in \{\underline{q}, \dots, \bar{q}\}$,

$$g_q(t) \triangleq \mathbb{E} \left[\varphi(q_T^*) + \int_t^T \ell(q_s^*) ds \right], \quad (\text{A.14})$$

and let $\mathbf{g}$ denote the collection of these functions, i.e., $\mathbf{g} = [g_{\bar{q}}(t), g_{\bar{q}-1}(t), \dots, g_{\underline{q}}(t)]'$. Then, $\mathbf{g} = \mathbf{V}\tilde{\mathbf{g}}$ where

$$\tilde{\mathbf{g}}_k(t) = \begin{cases} (\mathbf{V}^{-1}\boldsymbol{\varphi})_1 + (T-t)(\mathbf{V}^{-1}\boldsymbol{\ell})_1, & k = 1, \\ (\mathbf{V}^{-1}\boldsymbol{\varphi})_k e^{\mathbf{d}_k(T-t)} + \frac{e^{\mathbf{d}_k(T-t)}-1}{\mathbf{d}_k}(\mathbf{V}^{-1}\boldsymbol{\ell})_k, & \text{otherwise.} \end{cases} \quad (\text{A.15})$$

PROOF. Clearly, $g_q(t) + \int_0^t \ell(q_s) ds$ is a martingale, and therefore g satisfies the coupled system of ODEs

$$\begin{cases} \partial_t g_q(t) + \lambda^+ e^{-\kappa^+ \delta^{*+}(q)} (g_{q-1}(t) - g_q(t)) + \lambda^- e^{-\kappa^- \delta^{*-}(q)} (g_{q+1}(t) - g_q(t)) + \ell_q = 0, \\ g_q(T) = \varphi(q), \end{cases} \quad (\text{A.16})$$

which can compactly be written as $\partial_t \tilde{\mathbf{g}}(t) + \mathbf{B}\tilde{\mathbf{g}}(t) + \boldsymbol{\ell} = 0$ subject to $\tilde{\mathbf{g}}_q(T) = \varphi(q)$. Diagonalizing, and integrating, treating the zero eigenvalue separately, leads to the stated result. $\square$

¹²Recall these are independent of time and depend only on q .

Appendix B. Proof of Theorem 4

Here we prove the verification Theorem for the value function in (7). Let $(\delta_s^\pm)_{t \leq s \leq T} \in \mathcal{A}$ denote an arbitrary admissible control, $\delta_s = (\delta_s^+, \delta_s^-)$ denote the pair of controls, $(X_s^{x, \delta})_{t \leq s \leq T}$ (solving (2)) and $(N_s^{\pm, \delta})_{t \leq s \leq T}$ denote the trader's wealth and counting process for filled limit sell/buy orders, respectively, obtained by following the control with $X_{t-} = x$. Proceeding in the usual manner, write the tentative value function as $\hat{H}(t, x, S, q) = x + q(S - \alpha q) + h(t, q)$ (here, we write $h(t, q) = h_q(t)$ to ease readability). Next, observe that

$$\begin{aligned} \int_t^T dX_s + \int_t^T S_{s-} dq_s &= \int_t^T \int_{\mathbb{R}_+ \times [\delta_s^+, +\infty)} \delta_s^+ \mu^+(d\mathbf{y}, ds) + \int_t^T \int_{\mathbb{R}_+ \times [\delta_s^-, +\infty)} \delta_s^- \mu^-(d\mathbf{y}, ds), \\ [S, q]_{T-} - [S, q]_t &= \int_t^T \int_{\mathbb{R}_+ \times [\delta_s^+, +\infty)} y_1 \mu^+(d\mathbf{y}, ds) - \int_t^T \int_{\mathbb{R}_+ \times [\delta_s^-, +\infty)} y_1 \mu^-(d\mathbf{y}, ds), \quad \text{and} \\ q_{T-}^2 - q_t^2 &= \int_t^T \int_{\mathbb{R}_+ \times [\delta_s^+, +\infty)} (1 - 2q_{s-}) \mu^+(d\mathbf{y}, ds) + \int_t^T \int_{\mathbb{R}_+ \times [\delta_s^-, +\infty)} (1 + 2q_{s-}) \mu^-(d\mathbf{y}, ds). \end{aligned}$$

Note that only $\delta_s^\pm$ appear in the above and not $\delta_{s-}^\pm$ since the class of admissible strategies are taken to be $\mathcal{F}$ -predictable. Also, recall that $\delta_s^\pm$ are bounded from below and $\delta_s^+ = +\infty$ ($\delta_s^- = +\infty$) whenever $q_s = \underline{q}$ ($q_s = \bar{q}$). Using the above together with Ito's lemma for Poisson random measures and the fact that $h(t, q)$ given by Proposition 2 is differentiable in t , we then have

$$\begin{aligned} \hat{H}(T, X_{T-}^{x, \delta}, S_{T-}, q_{T-}^\delta) &= \hat{H}(t, x, S, q) + \int_t^T \partial_t h(s, q_s^{q, \delta}) ds \\ &\quad + \sigma \int_t^T q_s dW_s + \int_t^T \int_{\mathbb{R}^2} q_{s-} y_1 (\mu^+(d\mathbf{y}, ds) - \mu^-(d\mathbf{y}, ds)) \\ &\quad + \int_t^T \int_{\mathbb{R} \times [\delta_s^+, +\infty)} \left[\delta_s^+ + y_1 - \alpha(1 - 2q_{s-}) + h(s, q_s^{q, \delta} - 1) - h(s, q_s^{q, \delta}) \right] \mu^+(d\mathbf{y}, ds) \\ &\quad + \int_t^T \int_{\mathbb{R} \times [\delta_s^-, +\infty)} \left[\delta_s^- - y_1 - \alpha(1 + 2q_{s-}) + h(s, q_s^{q, \delta} + 1) - h(s, q_s^{q, \delta}) \right] \mu^-(d\mathbf{y}, ds). \end{aligned} \tag{B.2}$$

Next, note that, since the processes $\delta^\pm$ are bounded below (say $-\Delta^\pm \leq \delta_s^\pm \forall s \in [0, T]$), then

$$\begin{aligned} &\mathbb{E} \left[\int_0^T \int_{\mathbb{R} \times [\delta_s^\pm, +\infty)} |\delta_s^\pm|^2 \nu^\pm(d\mathbf{y}, ds) \right] \\ &\leq (\Delta^\pm)^2 \mathbb{E} \left[\int_0^T \int_{\mathbb{R} \times [-\Delta^\pm, +\infty)} \mathbb{I}_{\{\delta_s^\pm < 0\}} \nu^\pm(d\mathbf{y}, ds) \right] + \mathbb{E} \left[\int_0^T \int_{\mathbb{R} \times [\delta_s^\pm, +\infty)} (\delta_s^\pm)^2 \mathbb{I}_{\{\delta_s^\pm \geq 0\}} \nu^\pm(d\mathbf{y}, ds) \right] \\ &\leq \left(\lambda(\Delta^\pm)^2 e^{\kappa^\pm \Delta^\pm} + \frac{4e^{-2\lambda}}{(\kappa^\pm)^2} \right) T < +\infty, \end{aligned} \tag{B.3}$$

Furthermore, the process q is bounded in the interval $[q, \bar{q}]$ and finally, from Proposition (2), $|h(s, q_s^{q, \delta})| < C$ for some $C < +\infty$, therefore from (B.2) we have

$$\begin{aligned}
& \mathbb{E}_{t,x,q,S} \left[\widehat{H}(T, X_{T-}^{x,\delta}, S_{T-}, q_{T-}^{q,\delta}) \right] \\
&= \widehat{H}(t, x, S, q) + \mathbb{E}_{t,x,q,S} \left[\int_t^T \left\{ \partial_t h(s, q_s^{q,\delta}) + q_{s-}(\varepsilon^+ - \varepsilon^-) \right. \right. \\
&\quad \left. \left. + \lambda^+(\delta_s^+ + \varepsilon^+ - \alpha(1 - 2q_{s-}) + h(s, q_{s-}^{q,\delta} - 1) - h(s, q_{s-}^{q,\delta})) \right. \right. \\
&\quad \left. \left. + \lambda^-(\delta_s^- + \varepsilon^+ - \alpha(1 + 2q_{s-}) + h(s, q_{s-}^{q,\delta} + 1) - h(s, q_{s-}^{q,\delta})) \right\} ds \right] \\
&\leq \widehat{H}(t, x, S, q) + \phi \sigma^2 \mathbb{E}_{t,x,q,S} \left[\int_t^T (q_s^{q,\delta})^2 ds \right], \quad \text{by (7).}
\end{aligned} \tag{B.4}$$

The above inequality holds for any admissible strategy $\delta \in \mathcal{A}$ and is binding for the optimal strategy δ^* . Furthermore,

$$\mathbb{E}_{t,x,q,S} \left[\widehat{H}(T, X_{T-}^{x,\delta}, S_{T-}, q_{T-}^{q,\delta}) \right] = \mathbb{E}_{t,x,q,S} \left[\widehat{H}(T, X_T^{x,\delta}, S_T, q_T^{q,\delta}) \right] = \mathbb{E}_{t,x,q,S} \left[X_T^{x,\delta} + q_T^{q,\delta} (S_T - \alpha q_T^{q,\delta}) \right].$$

Hence we have,

$$\begin{aligned}
& \sup_{\delta \in \mathcal{A}} \mathbb{E}_{t,x,q,S} \left[X_T^{x,\delta} + q_T^{q,\delta} (S_T - \alpha q_T^{q,\delta}) - \phi \sigma^2 \int_t^T (q_s^{q,\delta})^2 ds \right] \\
&\leq \widehat{H}(t, x, S, q) = \mathbb{E}_{t,x,q,S} \left[X_T^{x,\delta^*} + q_T^{q,\delta^*} (S_T - \alpha q_T^{q,\delta^*}) - \phi \sigma^2 \int_t^T (q_s^{q,\delta^*})^2 ds \right]
\end{aligned} \tag{B.5}$$

which proves that $\widehat{H}(t, x, S, q)$ is indeed the value function of the original control problem (8). Moreover, the feedback controls given by (8) are clearly admissible. $\square$